Computational Power of Random Quantum Circuits in Arbitrary Geometries

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Empirical evidence for a gap between the computational powers of classical and quantum computers has been provided by experiments that sample the output distributions of two-dimensional quantum circuits. Many attempts to close this gap have utilized classical simulations based on tensor network techniques, and their limitations shed light on the improvements to quantum hardware required to frustrate classical simulability. In particular, quantum computers having in excess of approximately 50 qubits are primarily vulnerable to classical simulation due to restrictions on their gate fidelity and their connectivity, the latter determining how many gates are required (and, therefore, how much infidelity is suffered) in generating highly entangled states. Here, we describe recent hardware upgrades to Quantinuum's H2 quantum computer, enabling it to operate on up to 56 qubits with arbitrary connectivity and 99.843(5)% two-qubit gate fidelity. We define a class of circuits with random geometries that become hard to classically simulate in very low depth and implement them utilizing the flexible connectivity of H2. A careful analysis demonstrating the fast saturation of classical simulation complexity with depth indicates that H2 can yield data well beyond the reach of state-of-the art classical simulation methods at unprecedented fidelities. We find that the considerable difficulty of classically simulating H2 is likely limited only by qubit number, demonstrating the promise and scalability of the quantum charge-coupled device architecture as continued progress is made toward building larger machines.

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I. INTRODUCTION

It is widely expected that the computational power of quantum computers will be fully realized only by using quantum error correction (QEC). However, a significant gap exists between the capabilities of *all* existing quantum computing architectures (e.g., trapped ions, superconducting qubits, neutral atoms, etc.) and the requirements to perform large-scale error-corrected algorithms. To close this gap, desirable physical properties for a quantum computing architecture can readily be identified: All else being equal, it is generally better to have more qubits, lower physical error rates, higher connectivity, and faster clock speed. An honest assessment of the state of the field—supported empirically by the sheer variety of technologies under development—is that no one architecture is uniformly superior to the others with respect to all of these properties, each having strengths that can plausibly compensate for its weaknesses. For example, recent research into low-density parity check codes [1] suggests that high connectivity may greatly reduce the qubit resource requirements for QEC, potentially alleviating challenges in

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manufacturability or scalability. Similarly, a slow clock speed may be compensated by advantages that translate into higher fidelity, since a quantum computer can then produce the correct answer to a given problem with a higher probability, requiring fewer trials to succeed.

As quantum computing technologies mature, many researchers have begun to explore whether quantum computational advantage might be obtainable *without* quantum error correction [2–5]. A minimal demonstration of the viability of this program was provided in the pioneering work of Ref. [2] on random circuit sampling (RCS), in which a quantum computer performed a (contrived) task that appeared to be infeasible to simulate on even the most powerful classical computers. It is interesting to note that the same desirable qualities for achieving large-scale error-corrected quantum computing—speed, connectivity, fidelity, and size—all play a fundamental role in determining the relative separation between the computational power of quantum computers and classical computers for RCS. For example, tensor-network methods based on compressed contraction allow classical computers to simulate quantum circuits with imperfect fidelity [6,7], with the fidelities achievable in simulation determined primarily by the connectivity of the circuit (higher connectivity generally leading to lower effective fidelity given fixed classical resources). Therefore, increasing a quantum computer's connectivity, fidelity, or both can strongly frustrate such simulation methods.

In this paper, we describe upgrades to Quantinuum's H2 trapped-ion quantum charge-coupled device (QCCD) quantum computer that enable it to operate with up to 56 qubits while maintaining arbitrary connectivity and improving upon the high two-qubit gate fidelities reported in Ref. [8] (Sec. II). As a test of the H2 quantum computer's capabilities, we implement RCS in randomly assigned geometries. In Sec. III, we briefly review random circuit sampling as a test of quantum advantage and discuss some of its known limitations. We then discuss the impact of geometry on the cost of exactly simulating quantum circuits and describe a class of random geometries for which the simulation cost is provably worst case (i.e., scales exponentially in the qubit number) even at constant circuit depth. In Sec. IV, we show that the combination of low depth requirements (afforded by arbitrary connectivity), together with the high gate fidelities achieved on the H2 quantum computer, enable sampling from classically intractable circuits in an unprecedented range of fidelity: Circuits deep enough to saturate the cost of classical simulation by exact tensor-network contraction (assuming no memory constraints) can be executed without a single error about 35% of the time. Unlike previous RCS demonstrations, in which circuits have been carefully defined in reference to the most performant achievable gates, our circuits are comprised of natural perfect entanglers (equivalent to control-Z gates up to single-qubit rotations) and Haar-random single-qubit gates. All circuits are run "full stack" with the default settings that would be applied to jobs submitted by any user of H2, without any special-purpose compilation or calibration. Section V then presents a more detailed estimate of the classical complexity of simulating the experiments in Sec. IV. Historically, the most significant loophole in the claim that RCS is classically hard in practice stems from the low circuit fidelities that have been achievable for circuits deep enough to become hard to simulate classically. We argue that the high fidelities achieved in this work firmly close this loophole. Section VI considers the impact of connectivity and fidelity on the achievable classical simulation difficulty of RCS experiments given fixed run-time. Our primary conclusion is that, even with 56 qubits, the computational power of H2 for RCS is strongly limited by qubit number and not fidelity or clock speed, with the implication that the separation of computational power between QCCD-based trapped-ion quantum computers and classical computers will continue to grow very rapidly as the qubit number continues to be scaled up.

II. THE H2 QUANTUM COMPUTER

The H2-series quantum computers are built around a racetrack-shaped surface-electrode trap, illustrated in Fig. 1. Similar to previous work [8], each qubit is encoded in the ${}^2S_{1/2}$ $|F = 0(1), m_F = 0\rangle \equiv |0(1)\rangle$ hyperfine states of a ${}^{171}\mathrm{Yb}^+$ ion, and we use ${}^{138}\mathrm{Ba}^+$ ions as sympathetic coolants. The execution of circuits with arbitrary connectivity is achieved via the operations illustrated in Fig. 1 and described in the caption. Each of the colored circles in Fig. 1 represents a Yb ion (qubit), which is stored in a two-ion crystal with a single Ba coolant ion (the crystal order alternating between Yb-Ba and Ba-Yb around the track). Prior to application of a two-qubit gate or the physical swapping of two qubits, the two crystals associated with those two qubits are merged into a single Yb-Ba-Ba-Yb or Ba-Yb-Yb-Ba crystal. Physical swaps are performed in both configurations, but two-qubit gates are performed only in the former, as it leads to a lower total gate error. An initial 32-qubit configuration of H2 was described in detail in Ref. [8], though the trap described in that work was designed to eventually support larger qubit numbers. Here, we describe the system performance following a recent operational upgrade to 56 qubits. The primary technical challenge in scaling from $N = 32$ to $N = 56$ was the development of transport waveforms for a larger number of potential wells. The new waveforms use the same basic primitives to rearrange qubits—linear transport, split or combine, and physical swap; see Fig. 1(e)—but the exact trajectories and spacing between potential wells have been reconfigured.

All quantum operations including state preparation and measurement (SPAM), single-qubit ($1Q$) gates, and

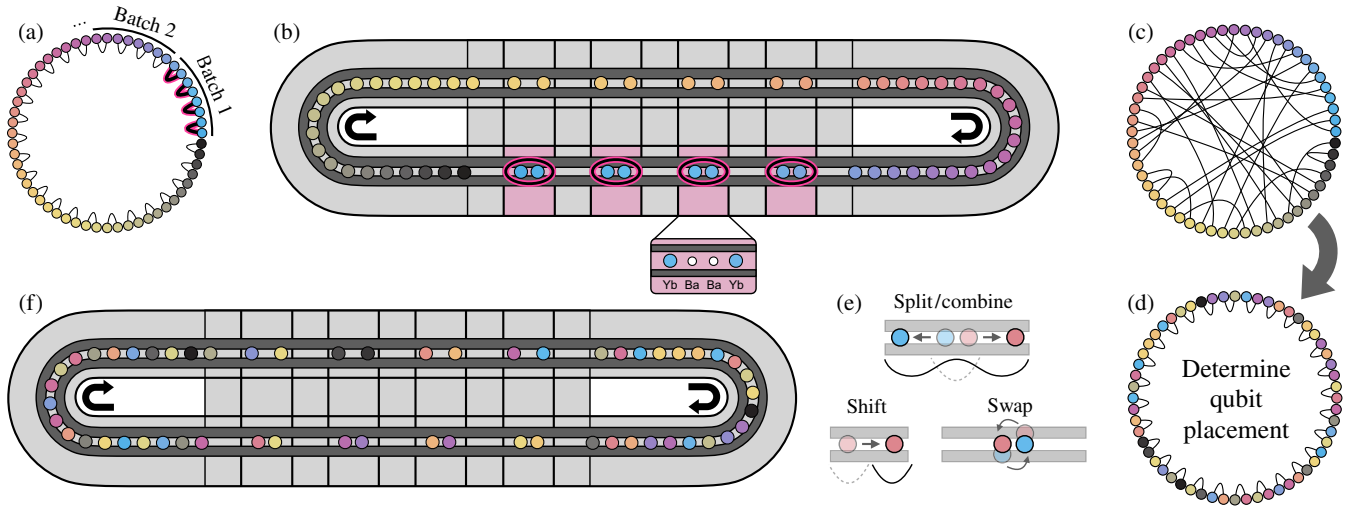


FIG. 1. Demonstration of how a densely gated circuit ($N/2$ $2Q$ gates per layer) with arbitrary connectivity is executed on the H2 quantum computer. (a) The first layer is executed by assigning each qubit to a unique $^{171}Yb^{+}$ ion (colored disk) such that gated qubits are collocated; each ion is labeled by a unique color, and the black lines connecting neighbors indicate that the associated qubits will be gated. Since H2 is currently configured with four active gate zones (see Ref. [8]), the two-qubit ($2Q$) gates are executed in seven batches of four parallel gates. (b) The first batch of four gates [highlighted in magenta in (a)] are executed in parallel in the bottom row of gate zones (the four pink-shaded regions), and qubits are then shuffled around the trap in a "rolodex" fashion until all gates (including $1Q$ gates) in the first layer have been applied. (c) The next layer of gates can act on arbitrary pairs. (d) An automated compilation step decides where to locate qubits in the trap (placement) and how to get them there (routing), resulting in a new assignment of qubit positions that ensures all pairs of qubits to be gated in this layer are once again collocated. Given the placement determined in (d), sequences of voltages are applied to the trap electrodes that, via a combination of the split or combine, shift, and swap operations shown in (e) achieve the desired ion placement for this layer of gates shown in (f). The gates can once again be executed in a rolodex fashion, and this entire process is repeated until all layers of gates have been applied. Note that additional coolant ions are omitted in these illustrations; as described in Refs. [8,9], each $^{171}Yb^{+}$ qubit ion is paired with a $^{138}Ba^{+}$ coolant ion.

two-qubit ($2Q$) gates are performed as described in Ref. [8] using only the bottom four (out of eight possible) gate zones [shaded pink regions in Fig. 1(b)]. The native $1Q$ gate and $2Q$ gate for all Quantinuum H-series hardware is the arbitrary-angle rotation $U_{1q}(\phi,\theta)=\exp[-i(\theta/2)(X\cos\phi+Y\sin\phi)]$ and the parametrized entangler $U_{ZZ}(\theta)=\exp[-i(\theta/2)Z\otimes Z]$, respectively. Both $1Q$ and $2Q$ gates are implemented using stimulated Raman transitions driven by pairs of lasers directed over the surface of the trap. Circuits described later in this manuscript are compiled into this gate set and then implemented verbatim (no additional gates, e.g., dynamical decoupling pulses, are applied).

Below, we describe in detail the characterization of errors for these different primitive operations, as well as characterization of errors incurred during the time spent between these operations. A summary of benchmarking results for these primitive operations is also given in Table I. We emphasize that the fidelity of primitive operations (gates, state preparation, and measurement) has not degraded relative to the initial $N=32$ configuration of H2. In fact, various technical improvements including reductions in laser phase noise have decreased the average error of our native perfect entangler from $\epsilon_{2Q}=1.83(5)\times10^{-3}$ measured in Ref. [8] to $\epsilon_{2Q}=1.57(5)\times10^{-3}$ reported in Table I and used to estimate circuit fidelities in Sec. IV. The primary detrimental impact of loading more qubits into the trap is that the average time to execute a layer of $N/2$ randomly selected gate pairs (depth-1 time) increased from 53 ms at $N=32$ to 80 ms at $N=56$, with an associated increase of the memory error per qubit per layer, ϵ_{mem}, from $2.2(3)\times10^{-4}$ at $N=32$ to $4.0(4)\times10^{-4}$ at $N=56$. After taking the data for this paper, additional improvements including the development of faster transport waveforms and the implementation of more efficient laser cooling methods have reduced the depth-1 time to around 70 ms, which together with other technical improvements to qubit frequency stability has

TABLE I. Average component benchmarking results for the tests described in Sec. II. All quantities labeled by subscripted ϵ are average infidelities. The value reported for p_{SPAM} is the mean of the probabilities $p(1|0)$ and $p(0|1)$, where $p(b|a)$ is the probability to measure b after preparing a qubit in state a.

Label	Test	Value ($\times10^{-4}$)
ϵ_{1Q}	$1Q$ RB	0.29(4)
ϵ_{2Q}	$2Q$ RB	15.7(5)
ϵ_{mem}	Transport $1Q$ RB	4.0(4)
p_{SPAM}	SPAM	14.7(9)

lowered the memory error per qubit per layer of gates to $\epsilon_{\text{mem}} = 2.5(5) \times 10^{-4}$ [10]. A combination of the improved laser cooling and an increased detuning of the $2Q$-gate lasers (to lower errors due to spontaneous photon scattering) have further reduced the $2Q$-gate error to the $\epsilon_{2Q} = 1.15(5) \times 10^{-3}$ reported in Ref. [10].

A. State preparation and measurement (SPAM) experiment

SPAM errors are measured with two experiments preparing all eight gate-zone qubits in the $|0\rangle$ (and $|1\rangle$) state and measuring how often the incorrect "1" (and "0") outputs are returned. The associated probabilities (the ratio between observed wrong outcomes and the number of trials) are denoted $p(1|0)$ and $p(0|1)$, respectively. Because of the nature of the photon counting measurement [11], there is an imbalance between probabilities $p(1|0) = 3.8(7) \times 10^{-4}$ and $p(0|1) = 26(2) \times 10^{-4}$. The average $[p(1|0) + p(0|1)]/2$ is reported in Table I as the SPAM error $p_{\text{SPAM}} = 1.47(9) \times 10^{-3}$. Uncertainty is calculated from the standard binomial confidence interval.

B. $1Q$ and $2Q$ randomized benchmarking ($1Q$ and $2Q$ RB)

$1Q$ randomized benchmarking ($1Q$ RB) was performed on the eight gate-zone qubits with standard Clifford randomization [12]. We used sequence lengths [2, 512, 2048] each with four randomizations and 100 shots per randomization as shown in Fig. 2(a). The RB decay rate λ_{1Q} is estimated by fitting the survival probability from each sequence length $p(m)$ with the first-order RB decay equation $p(m) = A\lambda_{1Q}^m + 1/2$ [12], where the asymptote is fixed to $1/2$ by randomizing the final output state [13]. The average $1Q$-gate infidelity is calculated by combining the results over all eight gate-zone qubits and converting the RB decay rate to an average infidelity per $1Q$ Clifford to give $\epsilon_{1Q} = (1 - \lambda_{1Q})/2 = 2.9(4) \times 10^{-5}$. Uncertainty is calculated from a semiparametric bootstrap resampling and shown with a one-sigma standard deviation [14,15].

$2Q$ randomized benchmarking ($2Q$ RB) was performed on the eight gate-zone qubits with standard Clifford randomization [12]. We used sequence lengths [2, 32, 128] each with four randomizations and 100 shots per randomization. $2Q$ RB was repeated seven total times interspersed over the entire data collection, and the results were combined and analyzed together as shown in Fig. 2(b). The RB decay rate λ_{2Q} is estimated by fitting the survival probability from each sequence length $p(m)$ with the first-order RB decay equation $p(m) = A\lambda_{2Q}^m + 1/4$ [12], where the asymptote is fixed to $1/4$ by randomizing the final output state [13]. The average $2Q$-gate infidelity was estimated by combining all four pairs of gate-zone qubit's $2Q$ RB data and scaling the average $2Q$ Clifford infidelity by the average number of $U_{ZZ}(\pi/2)$ gates per $2Q$ Clifford (1.5) to give $\epsilon_{2Q} = 3(1 - \lambda_{2Q}^{2/3})/4 = 1.57(5) \times 10^{-3}$. Uncertainty is calculated from a semiparametric bootstrap resampling and shown with a one-sigma standard deviation [14,15].

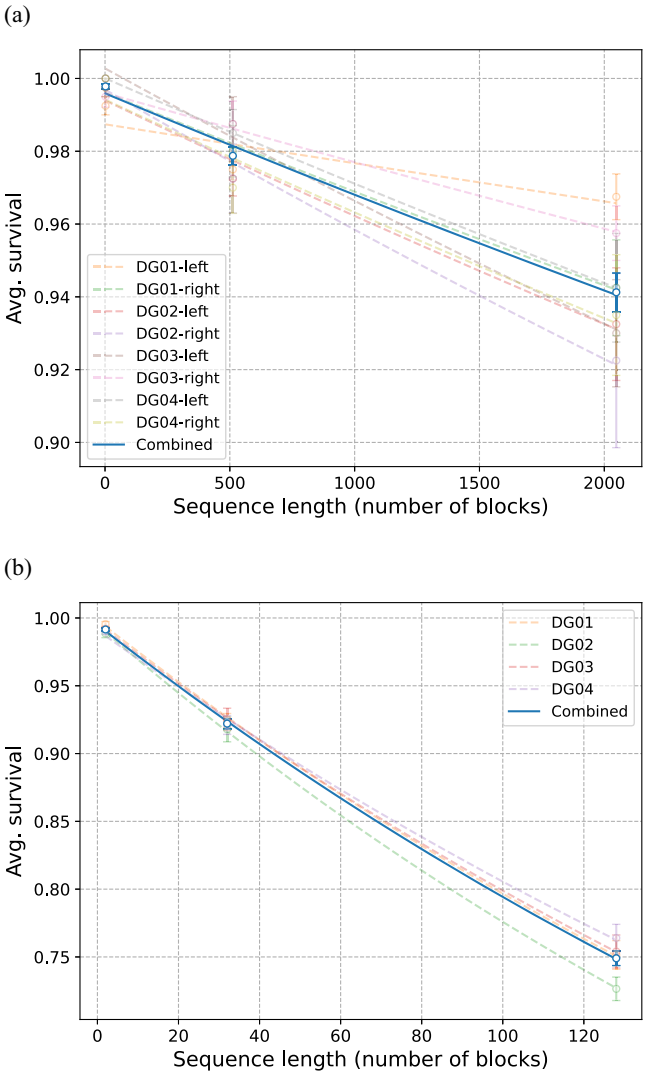


FIG. 2. (a) Experimental $1Q$ RB decay curve for all four gate zones (labeled DG01–DG04 in the legend) each with two qubits (dashed lines, the two qubits labeled as "left" and "right") and combined estimate averaged over all eight gate-zone qubits (solid line). (b) Experimental $2Q$ RB decay curve for all four gate zones (labeled DG01–DG04 in the legend, dashed lines) and combined estimate averaged over all four gate zones (solid line).

C. Transport one-qubit randomized benchmarking (transport $1Q$ RB)

Memory error refers to the cumulative effect of errors that happen to qubits when not doing $2Q$ gates. This memory error has several origins including reference clock and magnetic field drifts, imperfections in spatial phase tracking and other calibration errors or drifts, leakage out of the qubit manifold, and transport failures; see Ref. [8] for further discussion. Since memory error is closely related to

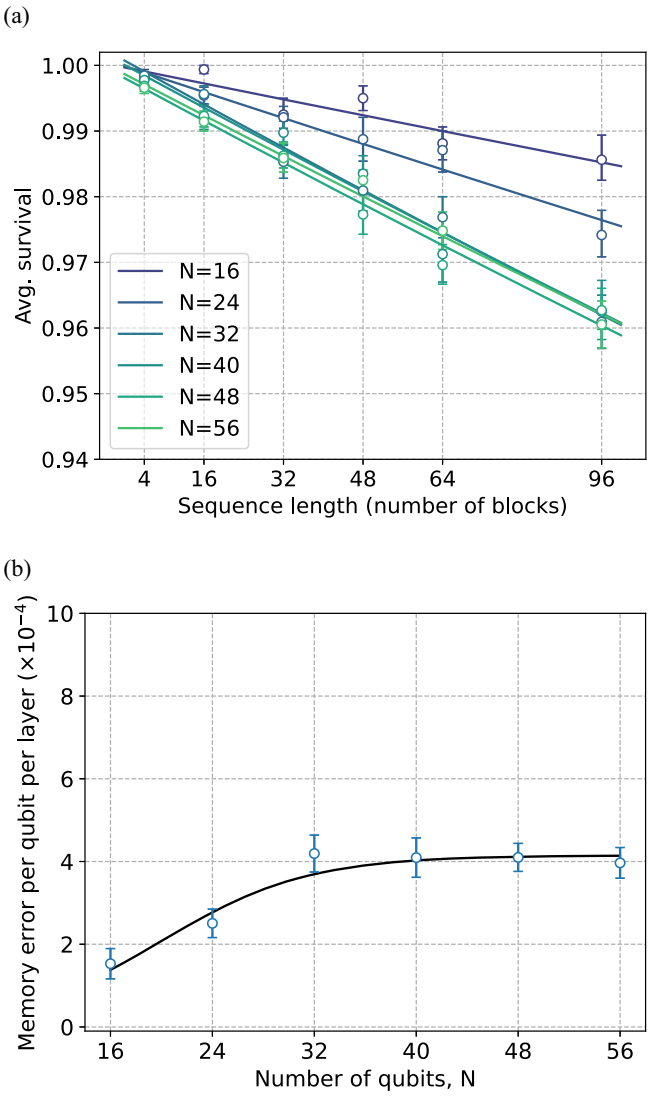


FIG. 3. (a) Experimental transport $1Q$ RB data and decay curves for each qubit number N at which the fidelity is estimated in the main text. (b) Memory error per qubit per layer extracted from the decay curves in (a), along with the logistic function model described in Appendix A.

the transport time, its magnitude varies with circuit geometry and qubit number. We quantify the memory error with transport $1Q$ randomized benchmarking experiments.

Transport $1Q$ randomized benchmarking (transport $1Q$ RB) was performed on up to all 56 qubits with random $1Q$ unitaries selected from SU(2). In transport $1Q$ RB, rounds of Haar-random $1Q$ random unitaries are interleaved with random transport operations. The random transport operations are selected to mimic the circuits with the same randomly assigned geometries used in the RCS experiments in Sec. IV and are implemented with a dummy $2Q$ gate that forces the ion transport and cooling but does not apply the lasers that generate the $2Q$ gate. We used sequence lengths $[4, 16, 32, 48, 64, 96]$ each with ten randomizations and ten shots per randomization, at each

TABLE II. Memory error per qubit per layer extracted from a separate transport $1Q$ RB experiment at each fixed qubit number N, and the resulting effective process infidelity per $2Q$ gate $\varepsilon(N)$.

Number of qubits	ϵ_{mem} $(\times 10^{-4})$	$\varepsilon(N)$: $(\times 10^{-3})$
$N = 16$	1.5(4)	2.4(2)
$N = 24$	2.5(4)	2.8(2)
$N = 32$	4.2(4)	3.1(2)
$N = 40$	4.1(5)	3.2(2)
$N = 48$	4.1(4)	3.2(1)
$N = 56$	4.0(4)	3.2(1)

$N \in [16, 24, 32, 40, 48, 56]$. The data for each qubit were accumulated and analyzed as a $1Q$ RB experiment (as detailed above) performed in parallel across all N qubits, as displayed in Fig. 3(a), allowing us to extract an error per qubit that accounts for memory errors accrued during the random transport as well as errors from the $1Q$ gates. In Table II, we report the resulting estimate of memory error per qubit per layer extracted from each experiment, which is also plotted in Fig. 3(b) along with an empirical model used for interpolation to compute ϵ_{mem} as a function of N, described in Appendix A. Uncertainty is calculated from a nonparametric bootstrap resampling with 1000 resamples and is shown with a one-sigma standard deviation.

As the qubit number is increased toward the maximum of 56 supported by the device, the time required for a full round of transport operations increases and eventually plateaus (since there are always 56 qubits in the trap, once a sufficient fraction of the qubits are involved in a given program, there is no longer a substantial increase in transport time with further increase in qubit number; see Fig. 13). We emphasize that the time reported for sorting at $N = 32$ is larger than that reported in Ref. [8] for $N = 32$, because the time required for $N = 32$ now includes transport operations on all 56 qubits (24 of which are loaded in the trap but not involved in the circuit).

Notably, the memory error reported in Table II is roughly a factor of 4 less than the $2Q$-gate error ϵ_{2Q} in H2. In Sec. IV, we discuss how to build a gate-counting model for circuit fidelity based on component fidelities; following that model, one can observe that the total contribution from memory error is somewhat less than that from the $2Q$ gate, although they are at roughly comparable scales. This should be contrasted with the overheads associated to implementation of an arbitrarily connected circuit on a 2D architecture, in which the required SWAP gates typically lead to a total error that is at least several times the native $2Q$-gate error, making implementation of arbitrarily connected circuits at scale implausible.

III. THE DIFFICULTY OF RANDOM CIRCUIT SAMPLING

Random circuit sampling (RCS) [16], as first envisioned and carried out on a superconducting quantum computer in

Ref. [2], is a computational task designed to highlight the difficulty of using a classical computer to simulate a quantum computer. There are four primary factors determining the separation between classical and quantum run times for RCS: the quantum computer's speed, size (number of qubits), gate fidelity, and connectivity. We note that, until very recently [17], existing work on random circuit sampling has been heavily focused on circuits with 2D connectivity [2,18–20]. The primary reason is that connectivity beyond 2D is technologically difficult to achieve in quantum computing platforms with fixed qubit locations, and to date 2D arrays of superconducting qubits are the only quantum computing architecture to have reached the system sizes and gate fidelities required to make RCS classically hard in practice (though significant progress toward executing computationally complex circuits with neutral atom arrays has recently been made [17]). Below, we show how improved connectivity specifically leads to dramatically more favorable scaling between quantum and classical run times for RCS.

A. Review of RCS

The task of RCS begins with a description of a random quantum circuit C, which we take to act on N qubits with d dense layers of $2Q$ gates ("dense" meaning that there are $N/2$ $2Q$ gates per layer). The quantum state $|\Psi\rangle$ obtained by applying C to the input state $|0\rangle^{\otimes N}$ induces a probability distribution over output bit strings x given by $P(x) = |\langle x|\Psi\rangle|^2$. Informally, the goal of RCS is simply to sample bit strings from a distribution that is suitably close to the ideal distribution P. Since RCS refers only to producing samples and not the actual state $|\Psi\rangle$, it can equally well be performed on a classical or quantum computer, making it a natural computational task for which their respective powers can be compared.

We first consider performing RCS with a quantum computer, which we assume can implement $2Q$ gates with an error probability (process infidelity) ε. For simplicity, in this preliminary discussion, we assume that all operations other than $2Q$ gates are perfect, though later calculations use a more sophisticated model that incorporates various measured sources of error into an *effective* per-$2Q$-gate process infidelity, as well as SPAM errors. The quantum computer executes the circuit M times and provides a list of M samples (length-N bit strings) $x_1, x_2, \ldots, x_M$, each drawn from the probability distribution $\mathscr{P}$ induced by the *noisy* circuit. One potential measure for the closeness of $\mathscr{P}$ to P—originally proposed in Ref. [2] but widely adopted in experimental RCS efforts [17–20] since then—is a suitably scaled and shifted version of the linear cross-entropy

$$F_{\mathrm{XEB}}^{\mathrm{lin}} = 2^N \left(\sum_x \mathscr{P}(x) P(x) \right) - 1 \tag{1}$$

$$\approx 2^N \left(\frac{1}{M} \sum_j P(x_j) \right) - 1, \tag{2}$$

where the second line is an estimator of the first using the bit strings sampled from $\mathscr{P}$. For sufficiently deep random quantum circuits, the probabilities P are themselves expected to be distributed according to the Porter-Thomas distribution. As a consequence, noiseless sampling [replacing $\mathscr{P} \to P$ in Eq. (1)] can readily be shown to result in $F_{\mathrm{XEB}}^{\mathrm{lin}} = 1$. On the other hand, if the circuit is so noisy that it outputs the completely mixed state, then the bit strings are sampled uniformly at random [$\mathscr{P}(x) = 1/2^N$], and Eq. (1) yields $F_{\mathrm{XEB}}^{\mathrm{lin}} = 0$.

While producing samples with a large $F_{\mathrm{XEB}}^{\mathrm{lin}}$ does not *necessarily* imply faithful sampling, the above two limits suggest that $F_{\mathrm{XEB}}^{\mathrm{lin}}$ might be a reasonable estimator of the fidelity F of the noisy state ρ output by the circuit, defined as $F = \langle\Psi|\rho|\Psi\rangle$. If so, then sampling from a quantum computer with a high $F_{\mathrm{XEB}}^{\mathrm{lin}}$ can serve as evidence of faithful RCS. A more careful justification for $F_{\mathrm{XEB}}^{\mathrm{lin}} \approx F$ can be made by treating the noise on the $2Q$ gates as stochastic Pauli error channels, such that with probability ε each $2Q$ gate is accompanied by a nonidentity Pauli error. For circuits that generate highly entangled states, one expects the presence of even a single Pauli error to cause both F and $F_{\mathrm{XEB}}^{\mathrm{lin}}$ to nearly vanish, in which case both must be approximately equal to the probability that no error has occurred in the entire circuit (up to small expected boundary effects [21]; see Appendix B):

$$F \approx F_{\mathrm{XEB}}^{\mathrm{lin}} \approx (1-\varepsilon)^{Nd/2}. \tag{3}$$

Early work on RCS generally assumed that any classical algorithm capable of producing samples with a nonzero $F_{\mathrm{XEB}}^{\mathrm{lin}}$ must essentially achieve a simulation of the quantum circuit in question with fidelity comparable to $F_{\mathrm{XEB}}^{\mathrm{lin}}$. Since simulating complex circuits with high fidelity is believed to be classically hard, this assumption makes it natural to *define* the task of RCS as simply the generation of samples with a particular nonzero value of $F_{\mathrm{XEB}}^{\mathrm{lin}}$. However, this definition has several potential drawbacks. First and foremost, computing $F_{\mathrm{XEB}}^{\mathrm{lin}}$ from quantum data in order to validate that RCS was performed is necessarily hard in the desired regime of quantum advantage, because computing the probabilities $P(x)$ of observed bit strings must be performed classically, and is at least as hard as classically sampling from the probability distribution P. Moreover, it is susceptible to the possibility that, while faithfully sampling a circuit's output distribution is classically hard [22–25], producing large $F_{\mathrm{XEB}}^{\mathrm{lin}}$ values *without* faithful sampling is not.

Several recent works have scrutinized the relationship between $F_{\mathrm{XEB}}^{\mathrm{lin}}$ and the output state fidelity F in

considerable detail [20,21,26]. Part of the conclusion of these works is that $F_{\mathrm{XEB}}^{\mathrm{lin}}$ is a good estimator of fidelity in the so-called "weak noise" regime, in which the probability of error per layer, $\varepsilon N/2$, is sufficiently small, and RCS experiments in the weak noise regime were carried out in Ref. [20]. When the noise rate exceeds a threshold value that depends on the choice of quantum gates and circuit geometry, there is a sharp phase transition into a "strong-noise" regime in which noise prevents quantum correlations from spreading across the system, and $F_{\mathrm{XEB}}^{\mathrm{lin}}$ can dramatically overestimate fidelity. The analysis in Ref. [26] for circuits with random geometries and the same gate set considered here, along with noisy simulations using a detailed error model for our machine, suggests that our experiments are very deeply in the "weak-noise" limit. Therefore, as long as circuits are deep enough for the output probabilities to be Porter-Thomas distributed, Eq. (3) is expected to hold (see Appendix B for further justification of this claim).

Reference [21] also points out that in a more adversarial setting, in which noise is tailored to make circuits easier to simulate by essentially removing gates in order to cut qubits into disconnected subsystems, particularly large discrepancies between $F_{\mathrm{XEB}}^{\mathrm{lin}}$ and F can be achieved in the limit that both are small. These results suggest a route to classically sampling with $F_{\mathrm{XEB}}^{\mathrm{lin}}$ values that are anomalously large compared to the achievable simulation fidelity, calling into question the use of achieving high linear cross-entropy scores *per se* as the definition of RCS. Reference [20] can be viewed as the first attempt to address such concerns by reducing noise rates in RCS experiments. The circuits executed on H2 and reported in this paper provide a substantially enhanced level of robustness against such classical attacks due to a combination of their high fidelity (which allows very few gates to be removed) and their high connectivity (which ensures that simulation remains difficult unless a large number of gates are removed). In particular, our circuits have their gates drawn from the edges of graphs with good edge-expansion properties (see Appendix E), which ultimately ensures that it is not possible to disconnect the system into two equally sized subregions without cutting an extensive number of gates, even at constant depth.

In light of the discussion above, in this work we primarily focus on providing direct evidence for the ability of the H2 quantum computer to sample with high fidelity from circuits that are difficult to classically simulate. Nevertheless, we do compute $F_{\mathrm{XEB}}^{\mathrm{lin}}$ for system sizes that we are able to ($N \leq 40$) and confirm that it agrees well with other fidelity estimates. We also report samples from circuits for which we are unable (due to the severe classical difficulty of simulating the circuits) to compute $F_{\mathrm{XEB}}^{\mathrm{lin}}$. We are generally unaware of *any* classical methods to produce $F_{\mathrm{XEB}}^{\mathrm{lin}}$ numbers comparable to what we expect from these circuits without performing essentially exact numerical simulations of the circuit (up to the modest loss in fidelity observed experimentally; see Sec. IV). While the above statement certainly warrants further scrutiny, we take this assumption as motivation to focus primarily on the difficulty of exact tensor network contraction in the remainder of this work. However, in Sec. V B, we also analyze the performance of tensor-network-based approximate simulation techniques and show that the random geometries explored here significantly increase (relative to local geometries) the classical resource requirements to achieve a given quality of approximation.

B. Impact of circuit geometry on exact simulation complexity

The most efficient known general-purpose classical method to simulate quantum circuits is to represent the probability of an output bit string as a tensor network (TN) and perform exact contraction of that network [27]. Together with rejection sampling or other related methods, such calculations enable classical RCS at a cost lower bounded by the cost of computing one such probability [28–30]. Actually contracting the TN associated to a single probability incurs a cost that can be *highly* dependent on the order in which the contraction of tensors is performed. For example, while there is always an ordering that leads to a timelike progression of the full state vector, with a computational cost that scales as 2^N, circuits of low depth and/or low connectivity generally admit contraction orders with much lower cost. Therefore, to assign a cost to the contraction of a given TN it is critical to minimize the contraction cost over all possible contraction orders. While it may be computationally hard to determine the optimal contraction order in general [27], many good heuristic methods have been developed in recent years. The contraction costs reported in this manuscript are obtained using cotengra [31], a PYTHON library that supports a variety of performant heuristic contraction order optimizers and hyperparameter optimization strategies to tune those optimizers. For the results discussed in this section, contraction order is optimized by targeting the minimization of floating-point operations (FLOPs) assuming no memory constraints. Since we have not performed exhaustive searches for FLOP-optimized contraction orderings, all reported costs are, strictly speaking, only upper bounds on the true contraction cost. However, it is also important to note that the assumption of unconstrained memory can cause one to seriously *underestimate* the cost of contraction in a more realistic memory-constrained setting. The deepest 56-qubit circuits run in this paper are likely *much* harder to contract in practice than the FLOP requirements computed in this section would suggest. In Sec. V A, we investigate the complexity of memory-constrained TN contraction for our circuits in more detail and show that it continues to grow extremely quickly with depth after the measure of complexity [see Eq. (6)] plotted in Fig. 5(c) saturates.

A convenient way to parametrize the contraction cost of the TN corresponding to a quantum circuit is to take the base-2 logarithm of the required FLOPs per gate for performing the contraction [32]:

$$\mathscr{N}_{d,N} = \log_2\left(\frac{\text{cost in FLOPs}}{Nd/2}\right). \tag{4}$$

Because the cost per gate (in FLOPs) of brute-force state vector simulation of $\mathscr{N}$ qubits scales as $2^{\mathscr{N}}$, $\mathscr{N}$ can be interpreted as an *effective qubit number* in the following sense: A depth d circuit C with cost $\mathscr{N}_{d,N}$ may act on $N > \mathscr{N}_{d,N}$ qubits, but it is only (roughly) as difficult to simulate as the worst-case hardness of simulating $\mathscr{N}_{d,N}$ qubits. At any fixed depth d, a circuit that is spatially local in D dimensions should have an effective qubit number scaling (asymptotically as $N \to \infty$) as the area of a minimal bisecting surface:

$$\mathscr{N}_{d,N} \sim d \times N^{(D-1)/D}. \tag{5}$$

This result is perhaps most recognizable in dimensions $D = 1, 2$, with the implication that the simulation cost of finite depth circuits is asymptotically independent of N for 1D circuits and exponential in $\sqrt{N}$ for 2D circuits. More generally, it follows from the existence of a spacelike contraction order in which, at intermediate steps of the contraction, the system is divided into two partitions having $O(d \times A)$ gates connecting them, with $A \sim N^{(D-1)/D}$ being the area of the hypersurface splitting the two partitions. It is also convenient to define a normalized effective qubit number:

$$\mathscr{C}_{d,N} \equiv \mathscr{N}_{d,N}/N \sim \frac{d}{N^{1/D}}, \tag{6}$$

which we refer to as the *complexity density* of the circuits. $\mathscr{C}_{d,N}$ quantifies what fraction of the qubits contributes to the worst-case complexity of simulation. Equation (6) implies that the asymptotic complexity density

$$\mathscr{C}_d \equiv \lim_{N\to\infty} \mathscr{C}_{d,N} \tag{7}$$

vanishes at any fixed depth for spatially local circuits, and, therefore, such circuits must be made deeper as they are made wider ($d \sim N$ in 1D or $d \sim \sqrt{N}$ in 2D) in order for the contribution of each qubit to the complexity of simulation to not vanish. To understand the behavior of very highly connected circuits, it is suggestive to consider the limit $D \to \infty$ in Eq. (6); evidently, the asymptotic complexity density will either be constant or decreasing more slowly with N than any power law. In this work, we consider a family of highly connected geometries for which $\mathscr{C}_d$ is provably bounded below by a constant (see Appendix E).

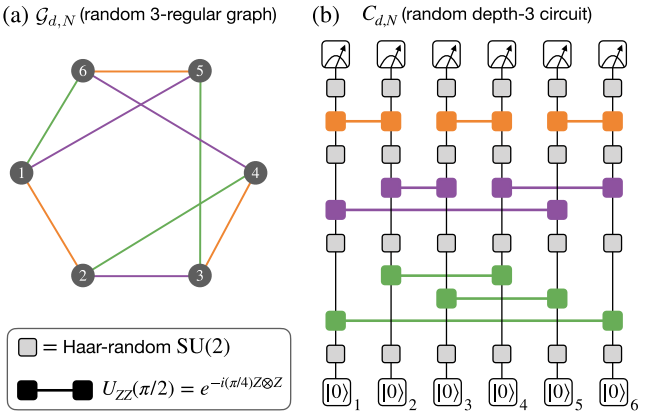


FIG. 4. The circuits considered in this paper have geometries induced by random regular graphs. For a depth-d circuit on N qubits, denoted $C_{d,N}$, we sample a random d-regular graph on N nodes, denoted $\mathcal{G}_{d,N}$ (the example shown above is for a depth-3 circuit on six qubits). To arrive at a circuit, we associate each vertex of $\mathcal{G}$ with a qubit in C and then assign $\mathcal{G}$ an edge coloring. Each color is associated with a layer of $2Q$ gates in C, with each edge of that color corresponding to a $2Q$ gate in that layer. Adjacent $2Q$ layers are separated by a layer of Haar-random $1Q$ gates on each qubit (with a single layer of $1Q$ gates immediately after state preparation and another immediately before measurement).

Given the flexibility of H2, we do not focus on a specific high-dimensional geometry but instead consider highly connected circuits with random geometries, a small example of which is shown in Fig. 4 (a detailed discussion of random-geometry circuit generation can be found in Appendix D). A random-geometry (RG) circuit $C_{d,N}$ of depth d on N qubits is generated by first selecting a random d-regular graph $\mathcal{G}_{d,N}$ with N nodes (d-regular meaning that exactly d edges impinge on each node). Each node of $\mathcal{G}_{d,N}$ is assigned to a qubit in $C_{d,N}$, and each edge to a $2Q$ gate.

Each of the $(N \times d/2)$ $2Q$ gates is chosen to be the perfect entangler $U_{ZZ}(\pi/2)$. The $2Q$ gates are then sorted into d layers with $N/2$ gates per layer by finding a proper edge coloring of $\mathcal{G}_{d,N}$ and then assigning one of the d colors to each of the d layers of the circuit. Edges of a given color then have their associated $2Q$ gates executed in the layer to which that color is assigned. A layer of Haar-random $1Q$ gates on every qubit is inserted immediately after qubit initialization, immediately before measurement, and between every layer of $2Q$ gates (for a total of $d+1$ $1Q$ layers). Each Haar-random $1Q$ gate is compiled into a single $U_{1q}(\phi,\theta)$ gate along with an arbitrary angle Z rotation, with the latter implemented in software via updates to the phase ϕ of each subsequent $U_{1q}(\phi,\theta)$ gate.

To highlight the impact of connectivity on contraction complexity, in Fig. 5 we compare the complexity density of RG circuits to that of 2D circuits. All complexity densities in this figure are derived [via Eqs. (4) and (6)] from contraction cost estimates obtained using cotengra and

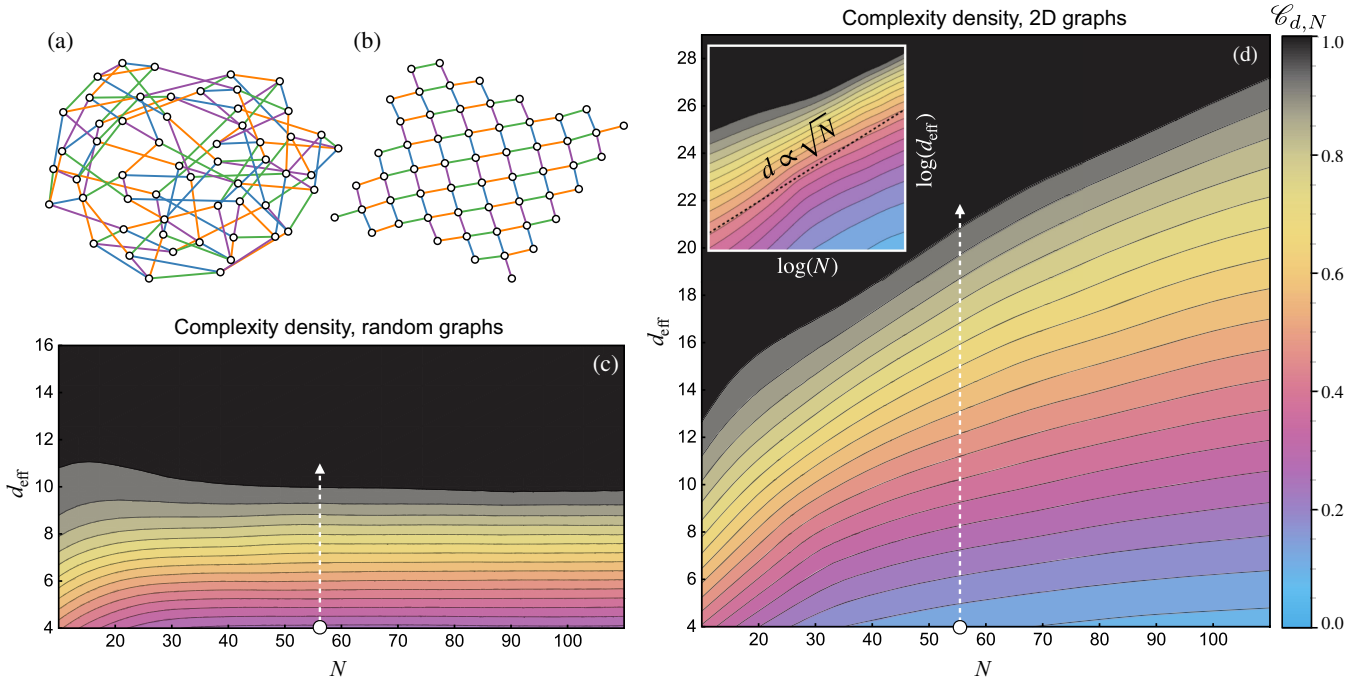


FIG. 5. Comparison of complexity density $\mathscr{C}_{d,N}$ for circuits with random geometries (RG) and 2D geometries. (a) and (b) show the gating pattern for a depth-4 graph on 56 qubits given random and 2D geometries, respectively. At each value (N, d), an N-node graph is assigned a coloring using either d colors (RG) or four colors (2D). Each circuit is comprised of layers of $U_{ZZ}(\pi/2)$ gates separated by layers of random SU(2) gates on each qubit, with the U_{ZZ} gates applied to each pair of qubits whose associated vertices are joined by an edge of one color. In (a) each layer corresponds to a unique color, while in (b) the four colors are repeated cyclically until the desired depth is reached. (c) and (d) show estimates of $\mathscr{C}_{d,N}$ for such RG and 2D circuits, respectively. While the complexity density for RG circuits is constant for $N \to \infty$ at fixed depth [evidenced by the horizontal contours in (c)], the log-log plot in the inset in (d) shows that $d \sim \sqrt{N}$ is required to maintain fixed complexity density in 2D. The white arrows in (c) and (d) highlight how far the depth needs to be increased at $N = 56$ before the complexity density saturates (i.e., the TN contraction cost becomes roughly equivalent to the cost of doing brute-force $N = 56$ state vector simulation).

targeting total FLOP minimization without any constraint on memory requirements (see Appendix H for a comparison of several state-of-the-art contraction-order optimizers). More realistic memory-constrained cost estimates, which are generally much higher for RG circuits for $d \gtrsim 12$, are reported in Sec. V A. The 2D circuits are similar in structure to the RG circuits described above, except that the gated pairs are drawn as edges of a 2D square lattice, as in Fig. 5(b). All gates corresponding to edges of a given color can be applied in a single layer, and for circuits with $d > 4$ layers we simply proceed cyclically through the color list until the desired depth is reached. To minimize contraction cost fluctuations due to boundary effects, we sample ten 2D grids at each N and d, where each grid is obtained by starting with a nominal square lattice with a vertex centered at the origin, applying a random offset and rotation to the lattice, and then selecting the N vertices with smallest Euclidean distance from the origin. The 2D circuits are slightly less densely gated than the RG circuits (which always have exactly $N/2$ gates per layer), because qubits at the edges of the 2D lattice are not gated in each layer. To make a fair comparison between RG and 2D circuits, we associate an effective circuit depth $d_{\text{eff}} = n_{2Q}/(N/2)$ to each circuit, with n_{2Q} the number of $2Q$ gates in the circuit (note that $d_{\text{eff}} = d$ for the densely gated RG circuits). Comparisons at fixed d_{eff} and N then always amount to comparisons at a fixed total number of $2Q$ gates. It is also important to note that, for this comparison, we have chosen the Quantinuum native $U_{ZZ}(\pi/2)$ gate as the $2Q$ gate. It is well known that iSWAP-like gates (gates with maximum entropy of their singular value decomposition) lead to relatively harder to simulate 2D circuits than $U_{ZZ}(\pi/2)$ when compared at fixed depth; the contraction cost estimates reported here for 2D circuits should not be conflated with direct cost estimates of circuits run in specific experiments on superconducting qubits, which will generally be higher for a given (N, d) given the use of rank-4 entangling gates. Nevertheless, the scaling arguments made here apply equally well regardless of the $2Q$-gate choice.

As discussed above, constant complexity density is maintained by scaling $d \sim \sqrt{N}$ in 2D [see the inset in Fig. 5(d)]. In contrast, RG circuits empirically achieve *fixed* complexity density at constant depth as $N \to \infty$, evidenced

by the flat contours in Fig. 5(c). The constant asymptotic complexity density observed in Fig. 5(c) is proven in the appendixes and is a consequence of the pairs of qubits to be gated in the circuit being chosen in correspondence with the edges of graphs with good asymptotic expansion properties (Appendix E). A sketch of the argument in that appendix is as follows: When a random circuit is generated from a d-regular graph, the TN representation of that circuit can be written as a network with similar topology to the underlying graph. The cost of contracting that tensor network can be lower bounded (i.e., by constant complexity density at large N) by showing that every contraction procedure has a step where an extensive $[\mathcal{O}(N)]$ number of legs leave the set of tensors contracted up to that step, so that the cost of contracting the network is at least $2^{\mathcal{O}(N)}$. The existence of such a step in the contraction procedure ultimately follows from the fact that d-regular graphs are one of the few known classes of "expander graphs" which possess exactly this property, phrased in terms of a bound on the "isoperimetric number." The white dashed arrows in Fig. 5 correspond to the circuits achievable in the current 56-qubit configuration of H2, and at this system size the depth at which $\mathscr{C}_{d,56}$ saturates to near unity (i.e., where the contraction cost saturates to the state vector simulation cost of 56 qubits) is roughly half as large ($d \approx 12$) for RG circuits as for 2D circuits ($d \approx 22$).

IV. THE FIDELITY OF RANDOM-GEOMETRY CIRCUITS ON H2

We have carried out RCS experiments on RG circuits at a variety of qubit numbers N and depths d, and all samples obtained from these experiments are reported in Ref. [33]. A small example RCS circuit ($N = 6$, $d = 3$) is shown in Fig. 6(a). The RCS experiments were randomly interleaved with other diagnostic experiments, including the $2Q$ RB and transport $1Q$ RB experiments in Sec. II, meant to validate their fidelity. Our primary tool for fidelity estimation is a mirror-benchmarking (MB) technique similar to

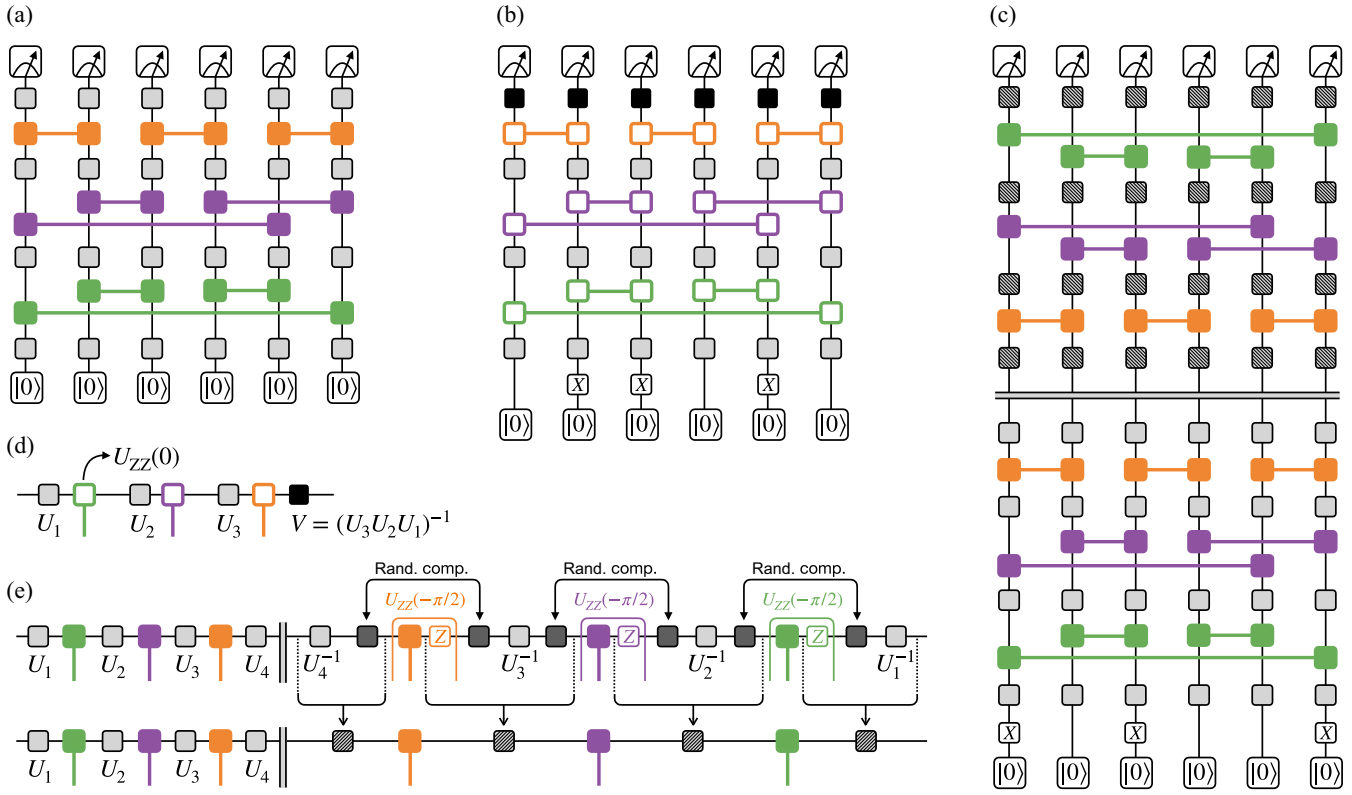


FIG. 6. Illustrative examples of the circuits implemented experimentally in this paper. (a) A depth-3 RCS circuit on six qubits (see Fig. 4 for a description of the notation). (b) Transport $1Q$ RB version of the same circuit. $2Q$ gates with unfilled boxes denote the use of a $U_{ZZ}(0)$ gate, which induces the same transport and cooling operations as the $U_{ZZ}(\pi/2)$ but does not apply a $2Q$ gate. The circuit is initialized in a random bit string, in this case 011010, and the $1Q$ gates at the end (solid black squares) invert the cumulative action of all prior $1Q$ gates in order to return to this initial state, as shown in (d) for a single qubit. (c) Mirrored version of the same circuit, here initialized in the random bit string 101001. On the reversed part of the circuit, all $1Q$ gates are inverted, $U_{ZZ}(-\pi/2)$ gates are created from $U_{ZZ}(\pi/2)$ gates by appending suitable $1Q$ Z rotations, and randomized compiling is used to twirl any potentially coherent errors in the $2Q$ gates into incoherent errors (avoiding any potential cancellation of coherent errors between the forward and backward halves of the circuit). All $1Q$ gates on the reverse half are then compiled down to the gates represented as hashed squares in (c), as shown in (e) for a single qubit.

that described in Refs. [34,35], which we carry out at all qubit numbers and circuit depths reported in Fig. 7. Mirror benchmarking provides a fidelity estimate for a depth-d circuit by mirroring a depth-$d/2$ circuit such that the output state is identical to the input state, as in Fig. 6(c). Randomized compiling [36] is used on the mirrored half of the circuit to prevent the potential cancellation of coherent errors. The randomized compiling is implemented by sandwiching each $2Q$-gate in the reverse half of the circuit by random $1Q$ Pauli gates such that the total action of the $2Q$ gate is left invariant, and additionally applying a Pauli Z operator to each qubit involved in the $2Q$ gate to convert the gate to positive angle, so that the same physical gate is consistently applied on both halves of the circuit. The additional $1Q$ gates introduced by this procedure are absorbed into the $1Q$ gates already applied between each entangling layer, corresponding to the hashed squares in Figs. 6(c) and 6(e). Bias in our SPAM fidelity is removed by preparing the initial state in a random bit string, such that the "correct" final measurement is an unbiased combination of 0 and 1 outcomes. The fraction of samples in which the qubits all return to the expected state, referred to as the MB return probability and denoted F_{MB}, serves as an estimate of the overall fidelity of the output state for a depth-d circuit. For all RCS circuits at $N \leq 40$, we also directly compute the linear cross-entropy $F_{\mathrm{XEB}}^{\mathrm{lin}}$ from the obtained samples, which serves as a complimentary estimate of the fidelity. For all higher qubit numbers—most notably, all data at $N = 56$—we still report samples from the RCS experiments in Ref. [33] but do not compute $F_{\mathrm{XEB}}^{\mathrm{lin}}$. While it should be computable for $N = 48$ with considerable effort, and at $N = 56$ for the shallowest depths, it does not appear possible to compute $F_{\mathrm{XEB}}^{\mathrm{lin}}$ for the $N = 56$ circuits at $d \gtrsim 12$ with any realistic amount of classical resources (see Sec. V A).

In addition to reporting the linear cross-entropy, we also provide the full (logarithmic) cross-entropy. Appropriately normalizing to the range [0, 1] amounts to subtracting out the log cross-entropy for the distribution of bit strings drawn uniformly from the Porter-Thomas distribution. The resulting quantity $F_{\mathrm{XEB}}^{\mathrm{log}}$ is closely related to the Kullback-Leibler (KL) divergence between the empirical ($\mathscr{P}$) and ideal (P) distributions of bit strings:

$$F_{\mathrm{XEB}}^{\mathrm{log}} = \gamma + N\log 2 + \sum_x \mathscr{P}(x)\log P(x) \tag{8}$$

$$\approx \gamma + N\log 2 + \frac{1}{M}\sum_j \log P(x_j). \tag{9}$$

Here, as in Sec. III, the second line is an estimator of the first using the M bit strings $\{x_j\}$ sampled from $\mathscr{P}$, and γ is Euler's constant. The uncertainty on the log cross-entropy is larger than that of the linear cross-entropy for $F \lesssim 0.32$, prompting the use of the latter in Ref. [2] where $F \ll 0.32$. By contrast, this experiment reports fidelities around that scale, for which there is no obstacle to use of the log cross-entropy, the more standard statistical distance measure.

We further employ a simple gate-counting estimate based on measured fidelities of the component operations in the circuit, referred to as the "digital error model" in Ref. [2]. The largest contribution to the overall circuit infidelity is $2Q$-gate errors. The $2Q$-gate average infidelity ϵ_{2Q} reported in Table I is inferred from the $2Q$ RB experiments that were interleaved with the rest of the data (see Sec. II B). The next leading source of circuit infidelity is memory error. We estimate the contribution of memory error using the separate transport $1Q$ RB experiments at each qubit number, whose circuits are depicted in Fig. 6(b) and results reported in Sec. II C and Table II. The geometry of the transport $1Q$ RB circuits at low depths is taken to exactly match the same circuits used in the RCS experiments. By making the replacement $U_{ZZ}(\pi/2) \to U_{ZZ}(0)$ for all $2Q$ gates, the transport $1Q$ RB circuits replicate nearly identical transport steps, cooling, $1Q$ gates, and overall timing as the RCS circuits without ever applying $2Q$ gates. In the absence of $2Q$ gates, the ideal circuit prepares each qubit in a known state. As described in Sec. II C, from the decay (with circuit depth) of the probability that each qubit ends in the correct state, we can extract an estimate of the aggregate non-$2Q$-gate contributions to the error per qubit per layer, denoted $\epsilon_{\mathrm{mem}}(N)$ [37].

Combining the memory error with the estimate of $2Q$-gate error from randomized benchmarking, the effective process or entanglement infidelity per $2Q$ gate is

$$\varepsilon(N) = \frac{5}{4}\epsilon_{2Q} + 2 \times \frac{3}{2}\epsilon_{\mathrm{mem}}(N), \tag{10}$$

where the prefactors incorporate the conversion between average and process fidelities [38] as well as the fact that there are two qubits in each $2Q$ gate [39]. Note that we do *not* include a contribution from ϵ_{1Q} listed in Table I, since $1Q$-gate errors are already accounted for in $\epsilon_{\mathrm{mem}}(N)$. A reasonable model for the overall fidelity of the circuit then simply counts the number of $2Q$ gates and state preparation and measurement operations and takes the corresponding fidelities to the appropriate power:

$$F_{\mathrm{GC}}(N,d) = \left(1-\varepsilon(N)\right)^{Nd/2}\left(1-p_{\mathrm{SPAM}}\right)^{N}. \tag{11}$$

The SPAM error p_{SPAM} is measured from an independent SPAM experiment and reported in Table I. As demonstrated in Appendix B, this gate-counting model for the circuit fidelity tends to systematically underestimate both the true circuit fidelity as well as the fidelity estimates $F_{\mathrm{XEB}}^{\mathrm{lin/log}}$ and F_{MB} provided by the cross-entropy and the average return probability of mirror circuits, respectively. Consequently, following the analysis in Appendix B, the effective depth in

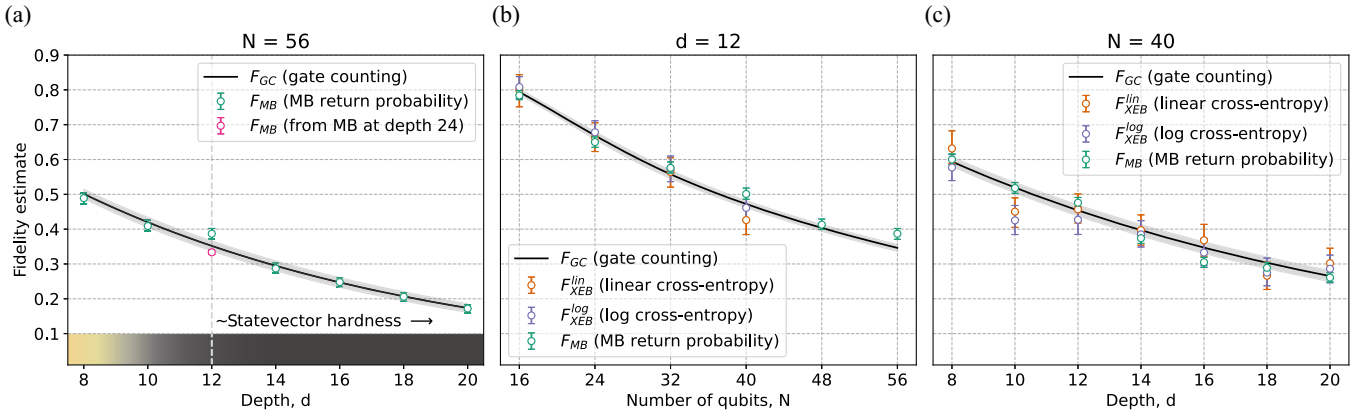


FIG. 7. Fidelity estimation from H2 experimental data at a variety of qubit numbers and circuit depths. (a) F_{MB} (MB return probability, green) at $N = 56$ as a function of depth. We also plot a fidelity estimate at depth 12 (pink) inferred from the MB return probability at depth 24 from mirror circuits that were constructed from the exact depth-12 RCS circuits. The color gradient corresponds to the complexity density from Fig. 5. (b) $F_{\mathrm{XEB}}^{\mathrm{lin}}$ (linear cross-entropy, orange), $F_{\mathrm{XEB}}^{\mathrm{log}}$ (log cross-entropy, purple), and F_{MB} (green) as a function of qubit number N at fixed depth $d = 12$. (c) $F_{\mathrm{XEB}}^{\mathrm{lin}}$ (orange), $F_{\mathrm{XEB}}^{\mathrm{log}}$ (purple), and F_{MB} (green) as a function of depth at $N = 40$, the largest of the N we took data at that was classically verifiable with our computational resources. All uncertainties plotted represent 1σ confidence intervals on the data. The gray shaded regions represent 1σ confidence intervals on the gate-counting model arising from propagation of uncertainties on the component operation fidelities.

the gate-counting model is shifted by approximately 1.12 for direct comparison with $F_{\mathrm{XEB}}^{\mathrm{lin}}$ and F_{MB}.

We report our experimental results in Fig. 7. The set of all circuits for which we report data in Figs. 7(a) and 7(b) were submitted to the H2 quantum computer in a random order so as to fairly average across any drift in computer performance, using default settings accessible to any user of the device. The set of $N = 40$ circuits for which we verify the linear cross-entropy $F_{\mathrm{XEB}}^{\mathrm{lin}}$ in Fig. 7(c) were submitted (and executed) immediately afterward, also in random order. Accompanying these $N = 40$ verification circuits, we performed a second transport $1Q$ RB experiment at $N = 40$ interleaved with the other circuits whose data are presented in Fig. 7(c). The results of this additional transport $1Q$ RB experiment are reported, along with more detailed circuit timing information, in Appendix A. In Fig. 7(a), where we expect that $F_{\mathrm{XEB}}^{\mathrm{lin}}$ is not computable except at the lowest depths with substantial effort, we report MB estimates of the fidelity achieved by H2 at 56 qubits on RG circuits out to depth 20. By depth 20, the analysis in Sec. V A suggests that, under reasonable assumptions about memory usage, classical simulations of these circuits are deeply infeasible. The estimated fidelities at these depths remain roughly 2 orders of magnitude larger than those reported in prior quantum supremacy claims based on RCS [2,18–20]. We also execute MB circuits at depth 24 that are mirrored versions of the exact depth-12 RCS circuits to provide another consistent estimate of the fidelity at depth 12 for $N = 56$. This procedure is similar to the one proposed in Ref. [40] but without the randomly compiled reference circuit and with SPAM errors added back in to better match expected circuit performance.

In Fig. 7(b), we show that $F_{\mathrm{XEB}}^{\mathrm{lin}}$ computed up to $N = 40$ agrees well with F_{MB} and the gate-counting model across a range of N at depth 12, where we expect that the cost of simulating the circuits is roughly equal to the state vector simulation cost; cf. Sec. III B. Further verification that $F_{\mathrm{XEB}}^{\mathrm{lin}}$ is well estimated by F_{MB} is provided by taking data at $N = 40$ across a wide range of circuit depths in Fig. 7(c) and computing $F_{\mathrm{XEB}}^{\mathrm{lin}}$ from the obtained samples on the Perlmutter supercomputer. All MB and RCS experiments were executed with 50 circuit randomizations and 20 shots per randomization at each depth, with the exception of the depth-24 experiment, which was executed with 50 circuit randomizations and 100 shots per randomization. All experimental data are provided in Ref. [33], including data from random circuits for which we could not compute $F_{\mathrm{XEB}}^{\mathrm{lin}}$.

V. EXACT AND APPROXIMATE SIMULATION COSTS

A. Exact contraction cost with memory constraints

Optimization of TN contraction solely targeting FLOP minimization can result in contraction orders that produce large intermediate tensors; the FLOP cost of a TN contraction is immaterial if those intermediate tensors cannot fit within available memory. While utilizing distributed storage for contractions is a theoretical possibility up to a point, in practice, all large-scale brute-force TN simulations of random quantum circuits [28–30] have favored a technique called *slicing* [41,42] that instead breaks the computation into many independent tasks, each able to fit onto a single GPU. It is, therefore, natural to optimize the

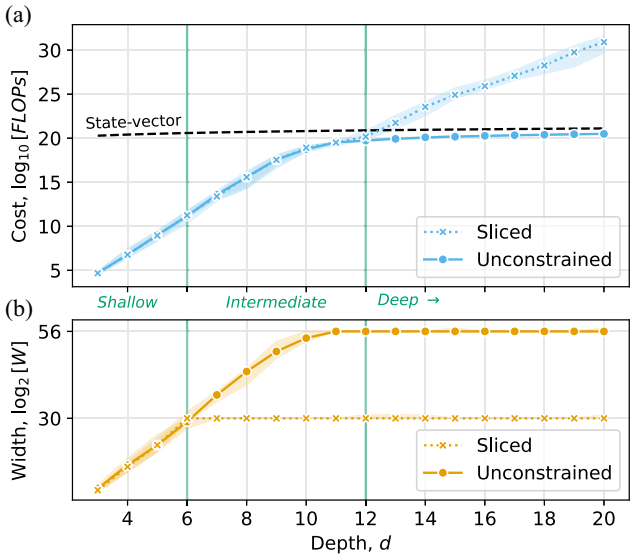


FIG. 8. The impact of constraining memory on cost for TN contraction of a single amplitude for random quantum circuits with $N = 56$. (a) FLOP cost of unconstrained optimized contraction paths compared with those sliced to $W = 2^{30}$ as a function of circuit depth d. The cost of state vector simulation is marked for reference. (b) Size of the largest intermediate tensor, or "contraction width," W, as a function of circuit depth. The lines represent the median behavior across 20 circuit instances, with the bands showing the min-max range.

FLOP cost of sliced contraction subject to a constraint on the memory footprint of each individual slice. The potentially enormous reduction in memory footprint is not always free, however, and at some point redundantly repeated operations introduce significant overhead. Below, we compare the memory-unconstrained and sliced cost for exact computation of a single amplitude ("strong simulation") of the random quantum circuits considered here, which serves as both a baseline for various tasks such as sampling and XEB verification and a useful comparison point to other circuits. We employ hypergraph partitioning [31,43] with simulated annealing refinement [29] to find highly optimized TN contraction paths, and in the sliced case enforce a maximum tensor size, or "width" $W = 2^{30}$, appropriate for the current generation of GPUs with 40–80 GB of memory.

In Figs. 8(a) and 8(b), we plot the contraction cost in FLOPs (assuming complex tensors) and the contraction width as a function of circuit depth d, averaged over 20 circuit instances at $N = 56$. We also show the reference FLOPs of state vector contraction and identify three distinct regions. First, a "shallow" region where both the cost and width increase exponentially but are still far below the state vector cost and no slicing is needed. Next, an "intermediate" regime where, even though the slicing reduces W by a factor of up to 2^{26}–64 million, it introduces no significant overhead. Finally, a "deep" regime where the unconstrained FLOPs and width converge with state vector, but the cost of the sliced contraction continues to grow exponentially.

At the intermediate to deep transition point around $d = 12$, where slicing overhead starts to emerge, the FLOP cost of approximately 10^{20} is already very significant, although certainly within reach of the largest supercomputers which can achieve approximately 10^{18} FLOPs per second [44]. However, the median sliced cost continues to exponentially grow to approximately 10^{30} as the depth increases to $d = 20$, putting the computation well out of reach. We note that, while improved techniques (that make use of extensive caching, for example) might bring this cost down somewhat, no such method is expected to fully remove the exponential cost of constraining $W \ll 2^N$ for deep circuits.

We also emphasize that these costs are meant primarily to situate the relative classical hardness of different depths with respect to other circuits. As discussed in Sec. VI, quantum RCS efforts generally produce S samples from C different circuits (for a total of $M = S \times C$ samples) such that the *expected* value of $F_{\mathrm{XEB}}^{\mathrm{lin}}$ averaged across all samples could, in principle, be distinguished from zero with high statistical confidence (putting aside the question of how *hard* the verification would be). To actually simulate the RCS experiments performed in this paper or to verify $F_{\mathrm{XEB}}^{\mathrm{lin}}$ of the samples generated by H2, one would want to perform at least S such amplitude contractions for each of the C circuits. While recent advances have shown that "multi-contraction" for a single circuit can be performed with cost sublinear in S [28–30], having to draw many samples still presents a significant computational increase. Moreover, in all RCS data that we report [33], we use a relatively small number of shots per circuit ($S = 20$), so the sublinearity of such methods provides only a marginal benefit. We also note that, when targeting $F_{\mathrm{XEB}}^{\mathrm{lin}} < 1$ in classical RCS, a generic speedup of $1/F_{\mathrm{XEB}}^{\mathrm{lin}}$ is available to brute-force TN methods [45]. Given the large fidelity values expected even for the deepest circuits implemented in this paper ($F \geq 0.1$ at $d = 24$; see Sec. IV), exploitation of hardware imperfections to perform classical RCS of our circuits at suitably reduced fidelity appears to offer no significant speedup.

B. The cost of approximate tensor network simulations

One promising method for approximately simulating quantum circuits is to utilize a tensor-network-based ansatz suitable for capturing states with limited bipartite entanglement. At sufficiently low circuit depths, when bipartite entanglement has not yet saturated to the Page limit (approximately $N/2$ for an equal bipartition) for a Haar random state [46], this method can be highly accurate or exact even for system sizes well beyond the reach of full state vector simulation. However, as the quantum state progresses through the circuit and becomes more entangled, it eventually needs to be compressed to maintain

an ansatz constrained by available resources (in both memory and run-time). This compression inevitably causes the state to lose fidelity with respect to the exact state, reminiscent of the way gate errors in a noisy quantum circuit cause a state to lose fidelity as the depth of a quantum circuit increases. The less noisy a quantum computer's gates are, and the more entanglement per gate that computer can generate, the more difficult it becomes for such methods to compete with the physically achieved fidelities for highly entangled states.

In Refs. [6,7], the authors provided strong analytical and numerical arguments that in the case of a matrix product state (MPS) ansatz and a random circuit, the overall fidelity F_{MPS} of the calculation is well approximated by simply accumulating the loss of fidelity in every compression step and is, therefore, readily accessible during the calculation despite the unavailability (in general) of the exact state. From the estimated simulation fidelity, one can ascribe an error per $2Q$ gate $\varepsilon_{\mathrm{MPS}}$ via the relation $F_{\mathrm{MPS}} = (1-\varepsilon_{\mathrm{MPS}})^{(\#\text{ of }2Q\text{ gates})}$. The per-gate error rate of the simulation—which can be systematically lowered at the expense of larger bond dimension (and, therefore, more classical resources)—can then be compared to the effective error per $2Q$ gate on quantum hardware, ε. Assuming that fidelity is well approximated by $F_{\mathrm{XEB}}^{\mathrm{lin}}$ for both the data output by a quantum computer and for this classical simulation method [47], quantum advantage in cross-entropy benchmarking is possible only if feasible classical resources cannot achieve large enough bond dimensions to maintain $\varepsilon_{\mathrm{MPS}} \lesssim \varepsilon$. Below, we show that the high gate fidelities and connectivity of H2 make simulations based on MPS extremely challenging and likely infeasible at the scale and fidelity of the circuits run in this work. It would be interesting to investigate to what extent approximate simulations could be improved by using a more general TN ansatz than the MPS considered here. However, fast (with circuit depth) suppression of *any* low-entanglement partitions is guaranteed by the way we construct our circuits. Ultimately, H2 can produce states that are near-maximally entangled with respect to all possible partitions while maintaining high state fidelities, and sampling from such states should pose substantial challenges for existing compression-based TN methods.

All MPS results reported in this section were obtained with a density-matrix renormalization group (DMRG) algorithm similar to that described in Ref. [7]. In particular, we approximate the amplitude of a particular output bit string of a circuit using a closed-simulation approach; e.g., we evolve one MPS forward from the initial bit string to the middle of the circuit, evolve another backward from the final bit string to the middle circuit, and then compute their overlap. We simulate only depth-20 random circuits and use intermediate MPS results after only $d/2 < 10$ forward and backward layers to estimate the achievable fidelity for circuits of all depths $d < 20$. Given the extremely nonlocal

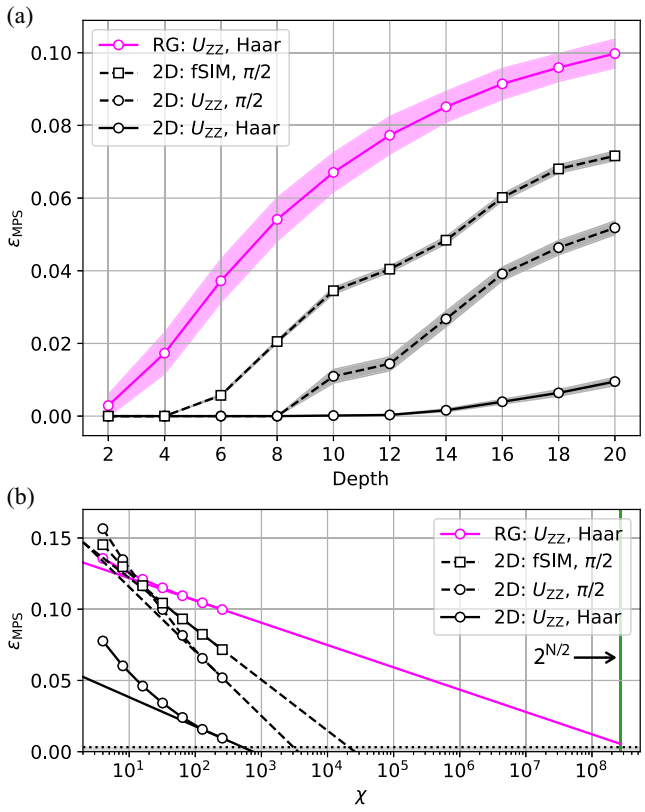


FIG. 9. (a) Achievable error per gate $\varepsilon_{\mathrm{MPS}}$ in simulation as a function of circuit depth using DMRG for circuits with 2D and random geometries and a variety of $1Q$- and $2Q$-gate sets. All curves use a bond dimension $\chi = 256$, and shaded regions show standard deviation of the error per gate across 100 (20) circuit randomizations for random (2D) circuits. (b) Error per gate at depth 20 as a function of bond dimension. Linear extrapolation of $\varepsilon_{\mathrm{MPS}}$ to the experimental value of $\varepsilon \approx 3.2 \times 10^{-3}$ (dotted horizontal line in this figure; see Sec. IV) from the two largest bond dimensions suggests that, by depth 20, DMRG cannot simulate random geometry circuits at the fidelities achieved on H2 without employing an essentially exact representation of the full state vector (MPS bond dimension $\chi = 2^{N/2=28}$, green vertical line).

geometry of our circuits, the use of a blocked MPS (in which many qubits are grouped together into a single tensor) can substantially reduce the number of long-range (interblock) gates applied during the time evolution if the blocks are chosen in reference to the specific random geometry in question. We explored a variety of blocking strategies using b blocks of $s = 56/b$ qubits per block (denoted $b \times [s]$), in each case making an assignment of qubits into blocks on a circuit-by-circuit basis by attempting to minimize the number of interblock gates using the open source hypergraph partitioning package KaHyPar [48,49].

Figure 9(a) shows how the error per gate grows as a function of circuit depth for both 2D (black curves) and RG (magenta curve) circuits and a variety of $1Q$- and $2Q$-gate sets. The label "fSIM" ("$\pi/2$") denotes the $2Q$ ($1Q$) gate set of Ref. [2], while "U_{ZZ}" ("Haar") denotes the $2Q$ ($1Q$) gate

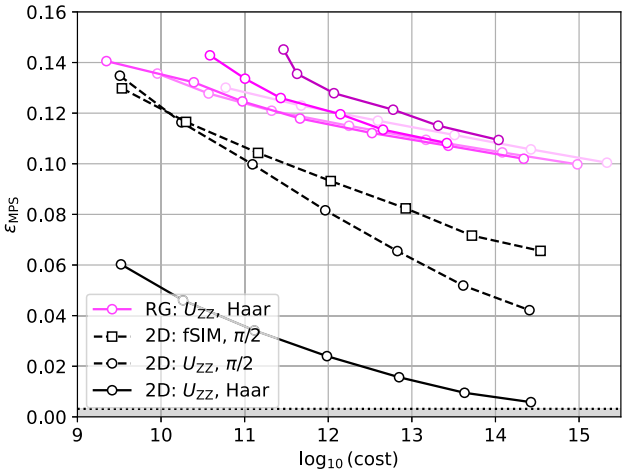


FIG. 10. Error per gate ($\varepsilon_{\mathrm{MPS}}$) versus computational cost (in FLOPs) for both 2D and RG circuits in simulation. The results for RG (2D) circuits are again averaged over 100 (20) random circuits. All 2D circuits employ a $7 \times [8]$ blocking strategy, while a variety of blocking strategies (each a different shade of pink; see the main text) are attempted for the RG circuits. Cost on the x axis is approximate and estimates the number of floating-point operations required for DMRG using various block sizes as the MPS bond dimension is increased (including the cost of applying all gates to the MPS and all MPS overlap calculations encountered in DMRG, but not the cost of QR decompositions).

set used for the circuits run on H2. For all 2D simulations, we used a $7 \times [8]$ blocking strategy on a grid similar to Google's Sycamore chip (though with a slightly different aspect ratio of seven columns and eight rows to accommodate 56 qubits, making each column an MPS tensor) and chose the same gating pattern as that used for the "quantum supremacy" circuits in Ref. [2]. For consistency, the RG circuit results are also plotted with a $7 \times [8]$ blocking strategy, though we show in Fig. 10 that results for random circuits are only weakly dependent on blocking strategy. Comparing the solid black and magenta curves, one can see that with the same gate sets RG circuits generate bipartite entanglement (heralded by the growing error rate at fixed bond dimension) much faster than 2D circuits. While utilizing $1Q$ and $2Q$ gates fine-tuned to generate maximum entanglement also can hinder the performance of DMRG in 2D, overall the impact of geometry is more severe. We note that the circuits run here could be made to entangle marginally faster by utilizing the $\pi/2$-labeled gate set in Fig. 9; see also the discussion in Appendix I. On the other hand, the $2Q$ fSIM gate utilized in Ref. [2] *does not* make RG circuits harder to simulate, because it can be decomposed as SWAP $\times\, U_{ZZ}(\theta)$; all SWAP gates in a given layer can be compiled into permutations of the qubit labels in future layers and result in a translation between different RG circuits in the same circuit ensemble considered. Figure 9(b) shows how the error per gate at depth 20 decreases when increasing the bond dimension χ of the MPS, which controls how much entanglement the MPS can capture. From a linear fit to the largest two bond dimensions ($\chi = 128$ and $\chi = 256$), one can see that by depth 20 the random circuits appear to require close to the maximum possible bond dimension of $2^{N/2}$ to achieve an error per gate comparable to the experiment [$\varepsilon(N = 56) \approx 3.2 \times 10^{-3}$, black dotted line], at which point the MPS ansatz becomes exact.

In Fig. 10, we compare the scaling of $\varepsilon_{\mathrm{MPS}}$ for RG circuits as a function of total computational cost using various blocking choices. Various shades of pink in Fig. 10 correspond to blocking choices $4 \times [14], 7 \times [8], 14 \times [4], 28 \times [2]$, and $56 \times [1]$ (ordered from light to dark). While the blocking choice significantly impacts the computational cost at fixed bond dimension, overall we find that different blocking choices are all roughly comparable from the standpoint of achievable $\varepsilon_{\mathrm{MPS}}$ at fixed computational cost. We also plot $\varepsilon_{\mathrm{MPS}}$ for 2D circuits of this geometry and using the same $1Q$- and $2Q$-gate set modifications as in Fig. 9. Once again, while the use of gate sets tailored to maximize the growth of entanglement can substantially increase the error per gate at fixed cost in 2D, the impact of random geometry on the cost to achieve a fixed error per gate is more severe.

VI. TIME REQUIREMENTS FOR VERIFIABLE RCS

Random circuit sampling—and any related statements about its asymptotic hardness, either classical or quantum—can be formulated purely in reference to the task of drawing samples from a particular distribution. There is a related but conceptually independent question regarding the sample efficiency of *verifying* random circuit sampling (or any sampling task): How many samples must be provided to offer compelling evidence of having sampled from the claimed distribution? If closeness to the ideal distribution is judged by $F_{\mathrm{XEB}}^{\mathrm{lin}}$, then the specific question becomes: If someone claims to be able to sample from a distribution with a particular value of $F_{\mathrm{XEB}}^{\mathrm{lin}}$, how many samples must they provide in order for us to be confident that they can, in fact, do so [50]? Since the trivial algorithm of guessing random bit strings produces $F_{\mathrm{XEB}}^{\mathrm{lin}} \approx 0$, the experimental task of RCS has often been implicitly defined as generating sufficiently many samples such that $F_{\mathrm{XEB}}^{\mathrm{lin}}$—were it ever to be computed from the provided samples—would be greater than zero with high statistical confidence.

Since the statistical uncertainty on $F_{\mathrm{XEB}}^{\mathrm{lin}}$ scales inversely with the square root of the number of samples taken, the total required quantum run time of a verifiable RCS experiment must scale as

$$T_q \sim \tau_q/(F_{\mathrm{XEB}}^{\mathrm{lin}})^2 = \tau_q/(1-\varepsilon)^{Nd} \approx \tau_q \times e^{\varepsilon N d}, \qquad (12)$$

where τ_q is the quantum run-time of a single shot of one circuit. This equation makes it clear that a very fast

quantum computer (small τ_q) may produce enough data to witness even small statistical deviations from complete noise or randomness ($F_{\mathrm{XEB}}^{\mathrm{lin}} \ll 1$) in a reasonable amount of time. On the other hand, increased fidelity and connectivity greatly enhance the deviations of a quantum computer's output from random, thereby reducing the amount of data required and potentially compensating for a slower clock speed. While the discussion of quantum run-time for verifiable RCS may seem academic, it is important to note that a similar trade-off between fidelity and clock speed can be identified in most efforts to stretch the power of noisy quantum computers via error mitigation techniques [51]. Ultimately, these methods amount to extracting a weak signal encoding the noiseless result of a calculation or simulation from a background of noise in a similar spirit to RCS. All else being equal, faster clock speeds are clearly advantageous. However, while individual circuit run times on H2 are slow compared to superconducting devices (about 1 s per sample for the circuits in this paper), the high fidelity and connectivity offset this deficit, enabling us to achieve a convincing demonstration of classically intractable RCS with comparable run times to other state-of-the-art demonstrations based on much faster architectures.

Equation (12) implicitly establishes a region of circuit volumes in the $N-d$ plane within which a quantum computer can extract verifiable RCS data given a fixed amount of total run-time T_q:

$$d \leq d_{\max}(N) \equiv \frac{\log(T_q/\tau_q)}{\varepsilon N}. \tag{13}$$

By computing $\mathcal{N}_{d,N}$ for circuits of a particular geometry and maximizing it over $d \leq d_{\max}(N)$, we can determine the maximum effective qubit number accessible to a given quantum computer in a fixed total amount of run-time T_q, denoted $\mathcal{N}_{\max}(\varepsilon, \tau_q, T_q)$, which, in turn, determines the maximum classical difficulty of simulating the data output by such an experiment. For this purpose, we compute simplified models of the contraction costs for 2D and RG circuits in order to extrapolate contraction costs beyond the range displayed in Fig. 5, with parameters fitted to the contraction cost estimates obtained from cotengra for $N \leq 110$ (see Appendix F). From these models, we can compare effective qubit numbers achievable in 2D and random geometries using the same random gate set over a broad range of assumptions about the circuit time τ_q and the $2Q$-gate error rate ε, as shown in Fig. 11 for a fixed total quantum run-time of $T_q = 100$ s. These results, like the calculation in Fig. 5 that they are based on, assume a rank-2 entangling gate for 2D and random geometries. In 2D geometries, using a rank-4 entangler (as in Ref. [2]) would slightly shift the 2D curves upward but does not change the qualitative behavior shown in Fig. 11.

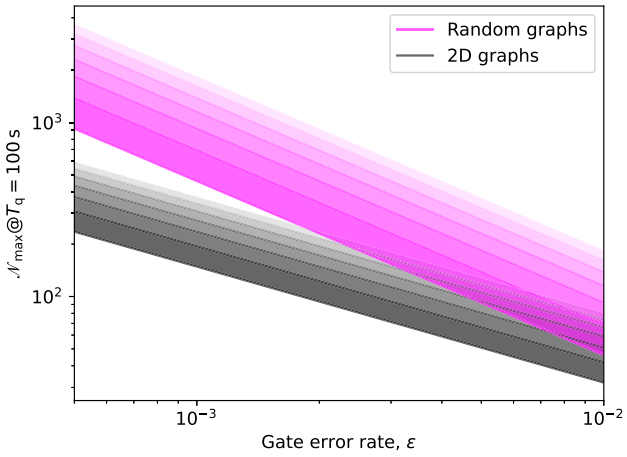


FIG. 11. Maximum effective qubit number $\mathcal{N}_{\max}$ theoretically achievable with 100 s of data as a function of the $2Q$-gate error rate. The black (magenta) solid curve at the bottom of each region shows results for 2D (random) geometries assuming a circuit execution time of $\tau_q = 1$ s. Each progressively fainter line above the lowest corresponds to another order-of-magnitude reduction in the assumed circuit time τ_q, with the uppermost dashed line corresponding to $\tau_q = 1$ μs for both geometries.

In this figure, the bottom boundary (darkest solid line) of each shaded region corresponds to an assumed circuit time of 1 s, and each progressively lighter line above represents a decrease of τ_q by one order of magnitude (with the topmost line corresponding to a circuit time of 1 μs). The fast saturation of classical simulation cost with depth for RG circuits translates into significant enhancements in the achievable effective qubit number (i.e., classical simulation costs) at fixed gate error and circuit time. We note that the estimated effective qubit numbers in Fig. 11 depend explicitly on the achievable error per gate ε. To what scale the QCCD architecture can be pushed while maintaining completely arbitrary connectivity at some fixed small ε is a technical question that will be answered only by technical progress. It is true that, at *some* system size, memory errors incurred during the transport required to implement RG circuits (which will scale with N in an architecture dependent way) will exceed the $2Q$-gate error rates. However, we believe the scale and current performance of H2 is far from the limits of what is achievable with current technologies, especially considering recent results on junction transport [52,53].

VII. OUTLOOK

Demonstrating a convincing computational advantage of quantum computers over classical computers is widely regarded as a key milestone on the path toward broadly useful quantum computation, and much progress toward this goal has been made in recent years. From a formal point of view, one would ideally like for such a demonstration to be based around a problem for which an

exponential asymptotic advantage can be claimed as the size of the problem (and the computer used to tackle it) grows. Since the original proposal for RCS as a task for demonstrating exponential quantum advantage [54], substantial progress has been made in understanding its asymptotic difficulty both with and without circuit noise. For example, compelling evidence now exists that *noiseless* RCS requires exponential (in the qubit number) run-time for a classical computer [22–24].

In the absence of quantum error correction, the asymptotics of *noisy* RCS (i.e., the task of sampling from the output distribution that results from running a quantum circuit with a fixed level of noise per gate) are relevant but remain comparatively less well understood. Notable recent developments include a proof that, under certain assumptions about the noise structure and circuit depth, RCS is asymptotically easy for classical computers if the error rate per gate is held constant as the system size is scaled up [55]. We emphasize that *none* of these results (hardness in the absence of noise or easiness at fixed noise rate) is conclusive regarding how the difficulty of RCS grows over a particular range of system sizes given a particular error rate. Easiness results suggest that, at sufficiently large noise rates, RCS may be classically feasible at all system sizes. But it is natural to expect that, as error rates are reduced, the system sizes over which the asymptotic difficulty of noiseless RCS controls the empirical hardness will grow [56].

Ultimately, which set of results controls the behavior of classical RCS in the range of system sizes and error rates accessible to experiments is unclear, but one thing is clear: The larger one can make a quantum computer while continuing to push down error rates (for example, in a manner that maintains overall circuit fidelities while increasing N), the larger the gap between classical and quantum run times will become. Since RCS is specifically designed to maximize this gap, it can be viewed as a lower bar for performing more structured (and useful) algorithms that are beyond the reach of classical simulation. Therefore, the further one can increase this gap for RCS, the more likely it is that we will find *useful* tasks at which non-error-corrected quantum computers dramatically outperform the best classical algorithms. The high gate fidelities and arbitrary connectivity afforded by the trapped-ion QCCD architecture have enabled RCS to be carried out in a computationally challenging regime and at completely unprecedented fidelities, leaving open considerable room to scale such demonstrations up *even without* further progress in reducing gate error rates. We are confident that H2 is by no means near the boundary of how far such demonstrations can be pushed using the QCCD architecture. For example, increasing system sizes while also reducing memory errors, circuit times, and $2Q$ error rates all appear achievable with current technologies by moving toward natively 2D trapping architectures [52,53], and switching to qubit ions (such as $^{137}Ba^{+}$) that afford better SPAM fidelities than $^{171}Yb^{+}$ [57] and admit visible wavelength laser-based $2Q$ gates with more favorable error budgets. Given that RG circuits require such low depths to become hard to simulate (and a fixed fraction of qubits contribute to the worst-case hardness of simulation even as N is scaled up at *constant* depth), near-term scaling progress in the QCCD architecture should enable the faithful execution of quantum circuits whose simulation lies well beyond the reach of any conceivable classical calculation.

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DATA AVAILABILITY

All data presented, along with OpenQASM descriptions of the circuits from which the data was measured, are available at [33].

APPENDIX A: ADDITIONAL MEMORY ERROR BENCHMARKING

A second transport $1Q$ RB experiment was also executed at $N = 40$ (see Fig. 12), interleaved with the verification set of $N = 40$ RCS circuits whose results are displayed in Fig. 7(c), since these circuits were executed subsequent to the rest. The memory error per qubit per layer for these circuits was measured to be $\epsilon_{\text{mem}} = 4.6(5) \times 10^{-4}$, corresponding to an effective process infidelity of $\varepsilon(40) = 3.4(1) \times 10^{-3}$.

In order to produce estimates of the fidelity from gate counting of circuits at intermediate qubit numbers in Fig. 7(b), we require a model for the memory error at all possible qubit numbers. As an empirical model for the resulting memory error as a function of N (interpolating between zero error as $N \to 0$ and a maximum level of error as $N \to 56$), we employ a logistic (sigmoid) function L of the form

$$L(N) = \frac{A}{1 + e^{-k(N-N_0)}}, \tag{A1}$$

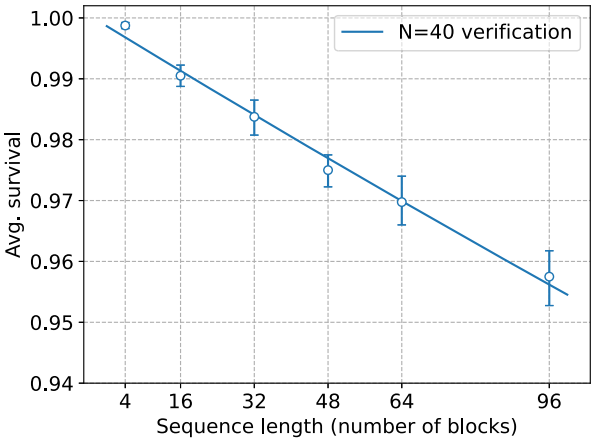


FIG. 12. Experimental transport $1Q$ RB data and decay curve for the $N = 40$ verification dataset collected subsequent to the rest of the data.

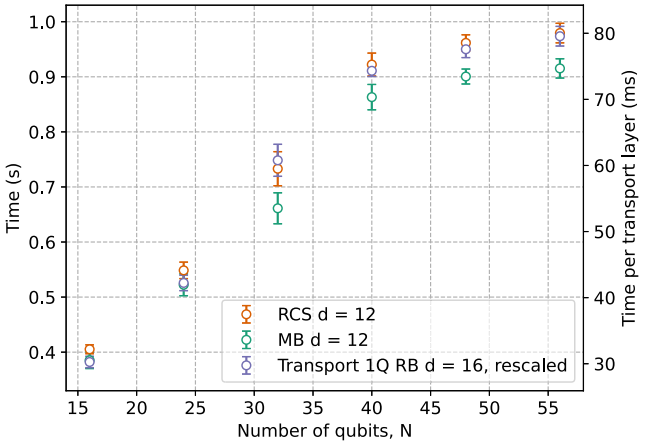


FIG. 13. Average time per shot on the H2 quantum computer (in experiment) for each type of circuit at $d = 12$ as a function of number of qubits, N. Since there is no transport $1Q$ RB data at sequence length 12, we take the transport $1Q$ RB data from sequence length 16 and rescale times linearly to length 12 (see Fig. 14 for evidence that the time per layer of transport $1Q$ RB sequences scales nearly perfectly linearly with depth).

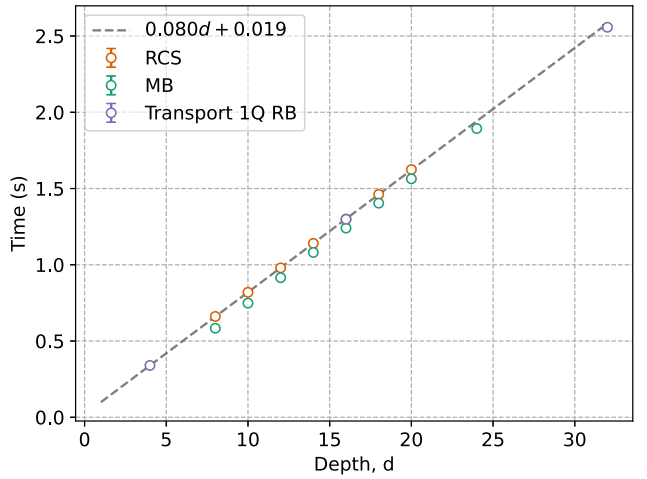


FIG. 14. Average time per shot on the H2 quantum computer (in experiment) for each type of circuit at $N = 56$ as a function of depth. For the transport $1Q$ RB (and RCS) data, the execution time is essentially exactly linear: fitting all depths out to 96 of the transport $1Q$ RB data yields the model: time $= (0.080d + 0.019)$ s, as depicted. Note that the range of the x axis is restricted to 32 in the plot to make it easier to visualize the MB and RCS data. Uncertainties are smaller than point size.

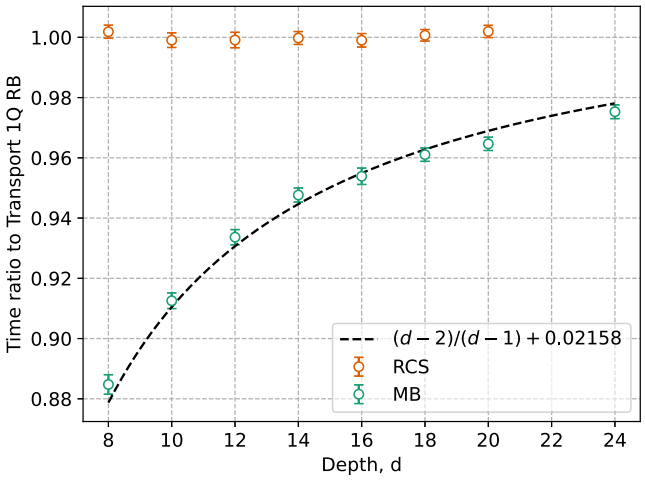


FIG. 15. The ratios of average shot time (in experiment) at each depth for $N = 56$ MB and RCS circuits to the expected time of transport $1Q$ RB circuits by interpolating the linear model in Fig. 14.

representing interpolation between extreme values of 0 and A, centered at N_0, with slope governed by k. A nonlinear least-squares fit of $L(N)$ to the measured memory error as a function of N yields the fit parameters $A = 4.1(2) \times 10^{-4}$, $N_0 = 20(2)$, and $k = 0.18(5)$, and is depicted in Fig. 3(b). Propagating these fit values and their uncertainties allows estimates of the memory error per qubit per layer at all N, continuously interpolating between $N = 16$ and $N = 56$. In combination with the other component benchmarks reported in Table I, the gate-counting model in Eq. (11) then supplies the estimates of the fidelity as a function of N reported as F_{GC} in Fig. 7(b).

It is interesting to examine to what extent shot time on the H2 quantum computer depends on circuit structure. In Fig. 14, we provide evidence that for RCS and transport $1Q$ RB circuits, which have identical gate topologies but different gate angles, the average time per shot at $N = 56$ is essentially identical and nearly perfectly linear with circuit depth. Mirror-benchmarking circuits, on the other hand, are expected to have one fewer layers of transport compared to RCS and transport $1Q$ RB circuits, since there should be no transport required to execute the first layer in the backward half of the circuit. In Fig. 15, we investigate the extent to which this is true by comparing the shot times of RCS and MB circuits to the times of the corresponding transport $1Q$ RB circuits at the same depth, computed by interpolating the linear model in Fig. 14. RCS circuits demonstrate essentially identical timings to the corresponding transport $1Q$ RB circuits. While MB circuits roughly demonstrate the expected $(d-2)/(d-1)$ scaling, they also have a constant multiplicative overhead of approximately 2.2% compared to transport $1Q$ RB that may be due to differences in the required transport to execute the mirror circuit.

APPENDIX B: COMPARISON OF VARIOUS FIDELITY ESTIMATORS

It is generally expected that the fidelity of a sufficiently deep random circuit will be roughly equal to the probability that no constituent operation has failed; we refer to this fidelity estimation strategy as gate counting and label the fidelity estimated in this manner F_{GC}. To estimate the actual fidelity of a circuit experimentally (in the regime where it is not computable even assuming an accurate noise model is known), we use the mirror-benchmarking technique

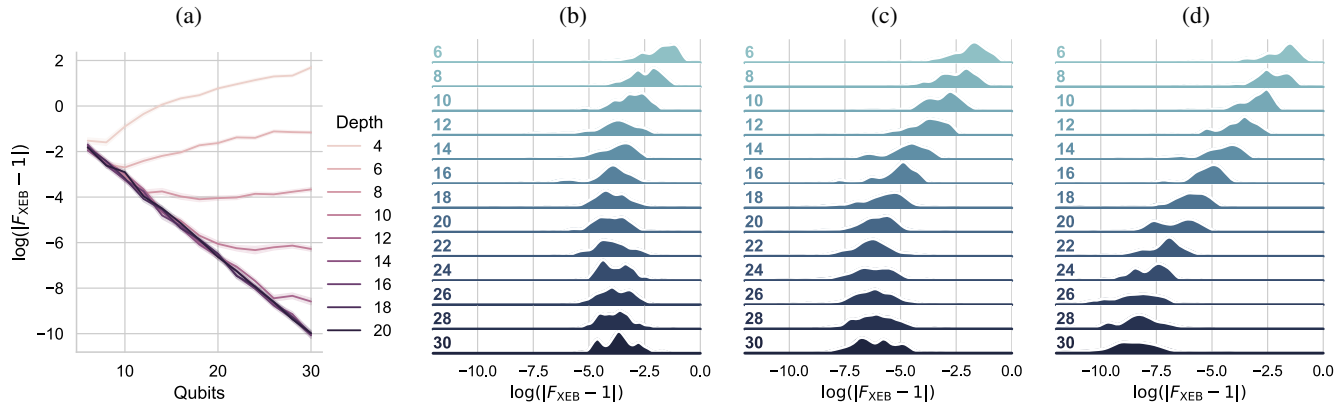


FIG. 16. (a) Deviations of $F^{\mathrm{lin}}_{\mathrm{XEB}}$ from unity in noiseless simulation as a function of qubit number N for RG circuits of various depths. Each curve shows the median over 100 circuits, and shaded regions represent one standard error. (b)–(d) Distributions of $|F^{\mathrm{lin}}_{\mathrm{XEB}} - 1|$ over circuits for various N at (b) $d = 8$, (c) $d = 10$, and (d) $d = 12$ in noiseless simulation. The histograms are smoothed probability densities of $\log|F^{\mathrm{lin}}_{\mathrm{XEB}} - 1|$ over 100 random circuits at the stated value of N (different N offset vertically, labeled on the left of each plot) and d. Small values indicate that the distribution of probabilities at the output of a given circuit is well converged (in its second moment) to the Porter-Thomas distribution. Over the limited range of N studied, it is clear that the vast majority of the distribution lies at values well below 1% for $d \geq 10$ and $N \geq 16$.

outlined in Ref. [34] adapted to the random circuits run here. In particular, to assess the fidelity of a depth-d random graph (RG) circuit, we take a depth-$d/2$ RG circuit (assuming d is even) and then append to it the same circuit with all gates inverted and run in the reverse order, such that the overall circuit (ideally) implements the identity map. Randomized compiling [36] is applied to the reverse circuits to prevent the possible cancellation of coherent errors. We choose the initial state to be a random computational basis state to remove any potential SPAM bias, measure the fraction of shots that return the (randomized) initial bit string, and denote this fraction F_{MB}. Strictly speaking, the decay of F_{MB} with depth provides an estimate of the variance of the Pauli fidelities of the per-layer error channel of the circuit. From this estimate, the per-layer fidelity can be bounded. However, as the numerics below confirm, F_{MB} provides an excellent estimate of the overall circuit fidelity even under a highly realistic local noise model.

For circuits deep enough that the output distribution of the ideal circuit is well approximated by the Porter-Thomas distribution, we also expect the linear cross-entropy fidelity $F^{\mathrm{lin}}_{\mathrm{XEB}}$ to agree with both of the above estimates. In this section, we analyze to what extent

$$F \approx F_{\mathrm{MB}} \approx F^{\mathrm{lin}}_{\mathrm{XEB}} \tag{B1}$$

is valid for the circuits in the manuscript. First, we assess the convergence to Porter-Thomas of the output distribution of ideal circuits as a function of depth and system size. At the depths where such convergence is adequate, we then run numerical simulations with a detailed noise model of our machine at a variety of moderate system sizes, checking that all three quantities agree in a manner that is very nearly system-size independent when the noise is scaled in such a manner that the probability of a gate failure per layer of $2Q$ gates is held fixed.

1. Convergence of RG circuits to Porter-Thomas without noise

For insufficient depths, the probabilities of bit strings at the output of RG circuits will not be distributed according to the Porter-Thomas distribution even in the absence of circuit noise, and linear XEB fails to be a good fidelity estimator. However, Fig. 16 shows that the convergence to Porter-Thomas in the second moment (equivalent to $|F^{\mathrm{lin}}_{\mathrm{XEB}} - 1| \to 0$ in the absence of noise) happens very quickly as the depth is increased, with the bulk of the distribution over 100 random circuits showing fluctuations well below the 1% level for $d \geq 10$. In Fig. 17, we show the full distribution of binned output probabilities at depths $d \in \{4, 6, 8, 10\}$. Even for $N = 16$ (the smallest number of qubits for which data is presented in the paper), upon averaging the distribution over 100 circuits we find that the convergence to Porter-Thomas is excellent over the entire

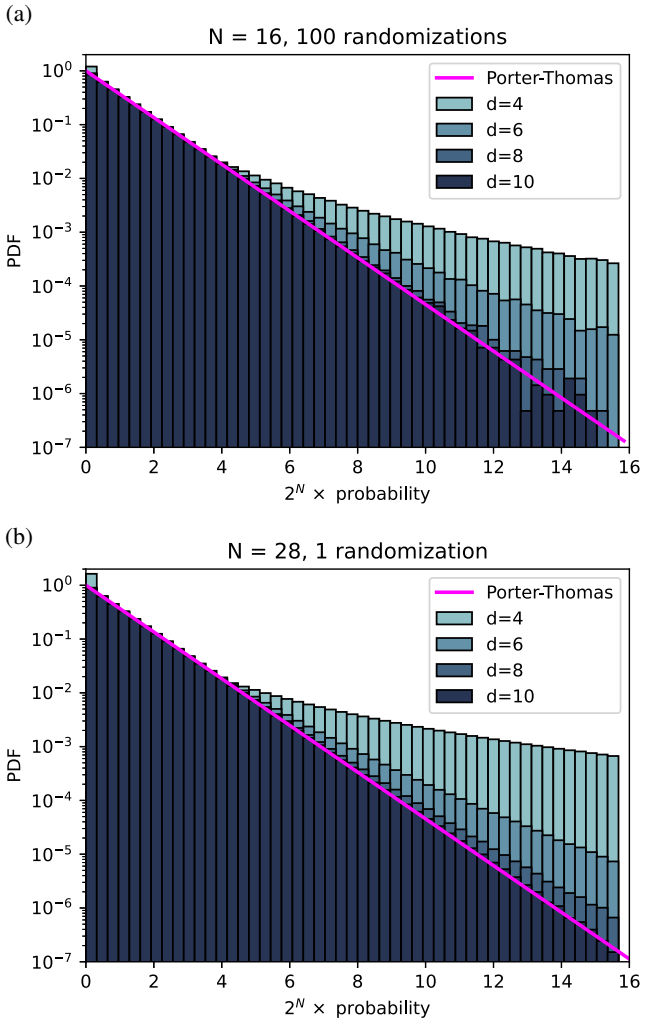


FIG. 17. Histograms of output probabilities for circuits on (a) $N = 16$ qubits and (b) $N = 24$ qubits, both at various depths in noiseless simulation. All probabilities (horizontal axis) are rescaled by 2^N, and the vertical axis is the normalized probability density function (PDF) over rescaled probabilities. By depth 10, the distribution is well converged to Porter-Thomas upon averaging over a 100 circuits for $N = 16$ and self-averages to Porter-Thomas for a single circuit by $N = 28$.

distribution by depth 10 [Fig. 17(a)]. By $N = 28$, we find good convergence at depth 10 even for a single random circuit [Fig. 17(b)].

2. Numerical studies with a detailed noise model

The simulations reported in this section utilize a noise model consisting of $2Q$ stochastic Pauli channels attached to each $U_{ZZ}(\pi/2)$ gate and a memory error attached to each qubit during the idling time between sequential $2Q$ gates that are applied to it. The stochastic Pauli channel is inferred from cycle benchmarking data [58], taken on H1-1 and rescaled according to the measured $2Q$ RB fidelity on H2 such that its average infidelity is the

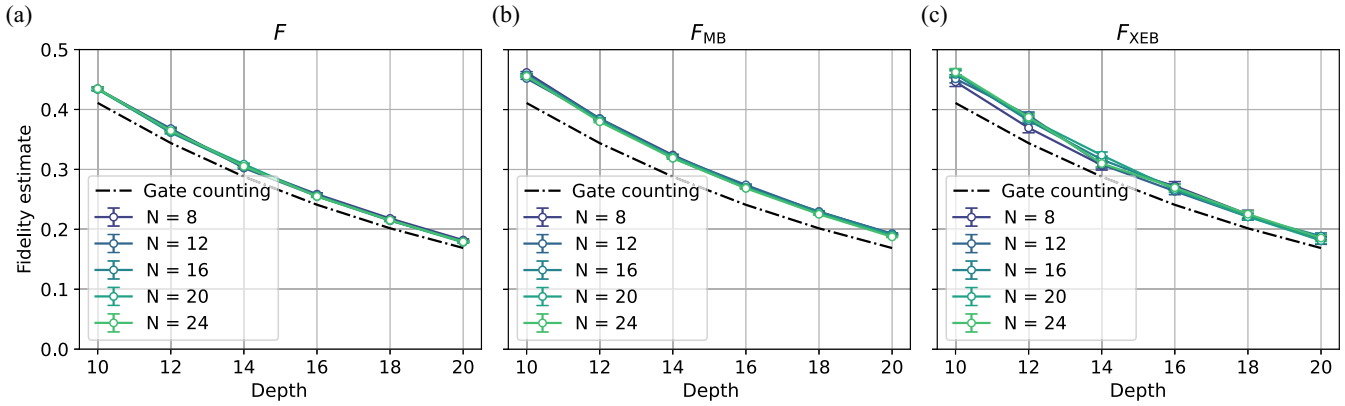


FIG. 18. Fidelity collapse when scanning N in noisy simulation. The three fidelity estimators plotted are (a) true fidelity of the output state with respect to the ideal state; (b) mirror-benchmarking fidelity, with the fidelity at depth d inferred from the return probability of a mirrored depth-$d/2$ circuit; (c) $F_{\mathrm{XEB}}^{\mathrm{lin}}$, the linear cross-entropy benchmarking fidelity. All simulations are randomized over 100 random circuits, and all error bars (generally too small to resolve) are 1σ confidence intervals from bootstrap resampling.

measured $\epsilon_{2Q} = 1.57 \times 10^{-3}$. The memory error per qubit per layer is inferred from transport $1Q$ RB data measured at $N = 56$ (see Sec. II C), and the associated average infidelity is $\epsilon_{\mathrm{mem}}(56) = 4.0 \times 10^{-4}$. While the transport $1Q$ RB data do not distinguish between different types of memory errors, we assume the per-qubit-per-layer memory error is entirely attributable to coherent dephasing; while we expect memory errors to be only partially coherent, we make this assumption because it seems most likely to challenge the assumptions underlying the various fidelity estimates. Finally, we rescale all noise rates by a factor of $56/N$ for simulations performed on N qubits, such that the error rate per layer of $2Q$ gates is held fixed to the $N = 56$ value as we change N. The gate-counting estimate works by converting these error rates into process infidelities and then multiplicatively composing the process fidelities of the circuit components as

$$F_{\mathrm{GC}} = \Big(1 - \varepsilon(N)\Big)^{Nd/2}, \tag{B2}$$

where here

$$\varepsilon(N) = \frac{56}{N}\left(\frac{5}{4}\epsilon_{2Q} + 2 \times \frac{3}{2}\epsilon_{\mathrm{mem}}(56)\right) \tag{B3}$$

is the rescaled effective process infidelity per $2Q$ gate. Figure 18(a) shows calculations of the actual circuit fidelity compared to this gate-counting estimate for various system sizes and over a range of circuit depths. The fidelities are obtained by quantum trajectories simulations, and we average over 100 random circuits taking 1024 trajectories for each. Error bars represent 1σ confidence intervals obtained by bootstrap resampling of the numerical data (1000 resamples). Figures 18(b) and 18(c) show simulations of F_{MB} and $F_{\mathrm{XEB}}^{\mathrm{lin}}$ under the same conditions (quantum trajectories simulations averaging over 100 circuits with 1024 shots per circuit, error bars from bootstrap resampling). An important conclusion of the numerical data in Fig. 18 is that the data collapse well for all N simulated, strongly suggesting that the observed similarity between the various fidelity estimates should hold at $N = 56$.

Figure 19 shows comparisons of F, F_{GC}, F_{MB}, and $F_{\mathrm{XEB}}^{\mathrm{lin}}$ at the largest simulated system size, $N = 24$. All three fidelity estimates agree reasonably well with the computed F, though there are small systematic differences between the various estimates that have a known physical origin [21]. The gate-counting argument assumes that if there are one or more errors in the circuit, the output state will be orthogonal to the ideal one. While this assumption is approximately true for very deep circuits, errors near the beginning of the circuit (when the state is only weakly entangled) do not effectively orthogonalize the state, and,

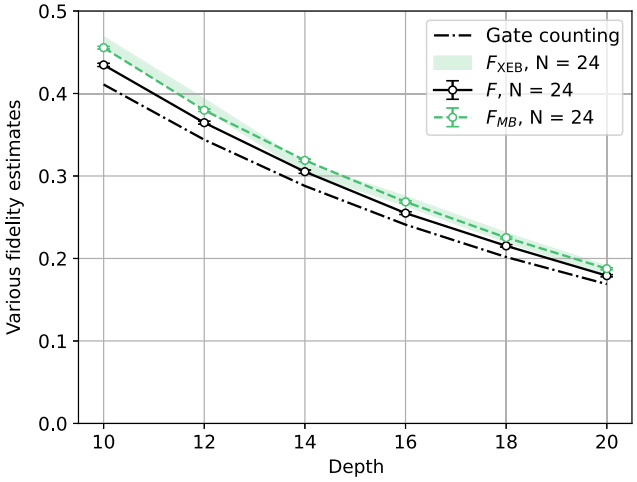


FIG. 19. Comparing various fidelities at $N = 24$, which was the largest system size we performed noisy simulations for. Error bars (for F and F_{MB}) and the shaded region (for $F_{\mathrm{XEB}}^{\mathrm{lin}}$) are 1σ confidence intervals obtained by bootstrap resampling.

therefore, gate counting underestimates the true fidelity. Both F_{MB} and $F_{\mathrm{XEB}}^{\mathrm{lin}}$ also reflect this imperfect state orthogonalization, but now due to errors at either the beginning *or* end of the circuit, causing these estimators to agree well with each other but to *overestimate* the true fidelity by the same amount that gate counting underestimates it. Note that the small discrepancies between these various fidelity estimates and the true fidelity are generally comparable to or smaller than the error bars reported in the manuscript, and we interpret the numerical data in Figs. 18 and 19 as confirmation that all three fidelity estimates—F_{GC}, F_{MB}, and $F_{\mathrm{XEB}}^{\mathrm{lin}}$—are suitably accurate for our purposes. In particular, these data support our expectation that $F_{\mathrm{XEB}}^{\mathrm{lin}}$ of the data taken from 56-qubit circuits (and available at Ref. [33]) should agree well with the mirror-benchmarking return probabilities for those circuits, despite the fact that we have not computed the cross-entropy of those data ourselves (and we do not know how it could be computed except possibly at the lowest depths with significant effort). However, because F_{GC} underestimates the true fidelity while $F_{\mathrm{MB/XEB}}$ overestimates it, the cumulative affect is a resolvable (at the level of our error bars) discrepancy between F_{GC} and $F_{\mathrm{MB/XEB}}$. Since the discrepancy arises due to the inadequate discounting of errors near the early- and late-time boundaries of the circuit by F_{GC}, we fit the gate-counting fidelity at a shifted depth $d \to d - \delta$ to the F_{MB} numerics at $N = 24$ to obtain a depth shift of $\delta \approx 1.12$. We use this inferred depth shift to report the gate-counting fidelities in Fig. 7 of the manuscript.

APPENDIX C: QUANTIFYING UNCERTAINTIES IN THE EXPERIMENTAL DATA

To compute the uncertainties associated to measurements of $F_{\mathrm{XEB}}^{\mathrm{lin}}$ and F_{MB}, we employ nonparametric bootstrapping [15]. An experimental dataset is a collection of observed bit strings $\{x_j\} = S$, and, for all $F_{\mathrm{XEB/MB}}$ data reported in Fig. 7, each data point corresponds to an experiment in which 50 random circuits are sampled 20 times each, for a total of 1000 samples per experiment. For a given outcome x_j, we let $f_j = 2^N P(x_j)$ [with $P(x_j)$ the probability of x_j for the ideal circuit] in the case of $F_{\mathrm{XEB}}^{\mathrm{lin}}$, or $f_j \in \{0, 1\}$ with "1" or "0" indicating a "right" or "wrong" value, respectively, in the case of F_{MB}. We let f be the expectation value of f_j, and $\hat{f}$ its estimate obtained by averaging f_j over the experimental outcomes: $\hat{f} = (1/N_s)\sum_j f_j$, where N_s is the total number of samples. In the nonparametric bootstrap procedure, one creates a large number r of synthetic datasets $\{\mathcal{S}_1, \mathcal{S}_2, \ldots, \mathcal{S}_r\}$, each one obtained by resampling the experimental data with replacement. The quantity of interest is then averaged over the data in each synthetic dataset to produce a distribution of estimates $\{\hat{f}_1, \hat{f}_2, \ldots \hat{f}_r\}$. A 1σ confidence interval $[f_{-1\sigma}, f_{+1\sigma}]$ is assigned to this distribution according to the appropriate 1σ quantiles, and a confidence interval for f is then constructed by reflecting the bootstrapped confidence interval about twice the empirically observed mean $\hat{f}$:

$$\mathrm{CI} = [2\hat{f} - f_{+1\sigma}, 2\hat{f} - f_{-1\sigma}]. \tag{C1}$$

This reflection is intended to remove bias in the construction of the bootstrap interval arising from the fact that the mean of the bootstrap distribution may not match the empirically observed mean.

We consider two methods for resampling from the observed dataset. One possible method is to pool together all of the samples associated to a particular experiment and resample from this combined pool, which we call the *aggregate resampling* method. Another, which we call the *double resampling* method, is to first resample from the pool of circuits associated to a particular experiment and then to further resample from the observed samples for each of the resampled circuits. In general, as we provide evidence for below, the double resampling method results in a more conservative (wider) confidence interval. To maintain consistency with previously reported methods for randomized benchmarking [8], we choose the double resampling method to quantify uncertainties of component benchmarks. However, which confidence interval is most appropriate depends on the extent to which the underlying distribution being sampled from for each circuit fluctuates from circuit to circuit. To understand which is more appropriate in the context of estimating F_{MB} and $F_{\mathrm{XEB}}^{\mathrm{lin}}$, we first assess to what extent these quantities vary from circuit to circuit given a fixed error model.

From numerics on small system sizes, we believe the fluctuations in both quantities from circuit to circuit are well below the 1% level for the circuits run in this paper ($d \geq 8$ at $N = 56$ and $d \geq 12$ at all $N \geq 16$). In the case of $F_{\mathrm{XEB}}^{\mathrm{lin}}$, evidence for this statement is provided in Fig. 20. There, we perform density matrix simulations of noisy circuits at small system sizes ($N \leq 14$) and observe that the standard deviation of $F_{\mathrm{XEB}}^{\mathrm{lin}}$ over circuits falls below 10^{-2} for all depths $d \geq 12$. A more relevant source of fluctuations from circuit to circuit comes from drift in machine performance. Considering that all circuits for all experiments are randomly interleaved in time, while all shots for a single circuit are taken in rapid succession, it is reasonable to model drifts in machine performance by assigning each circuit a random but fixed value of ε, drawn from a distribution around its experimentally measured average. For $F_{\mathrm{XEB}}^{\mathrm{lin}}$, we model a single circuit as sampling from a model probability distribution that is a convex mixture of the ideal Porter-Thomas distribution [with probability $F(\varepsilon)$] and the uniform distribution [with probability $1 - F(\varepsilon)$]. For F_{MB}, we model each circuit as a Bernoulli trial (biased coin flip) with a probability $F(\varepsilon)$ of returning the correct bit string and probability $1 - F(\varepsilon)$ of a wrong bit string. In both cases, the circuit fidelity F depends on the per $2Q$-gate process infidelity ε, which we

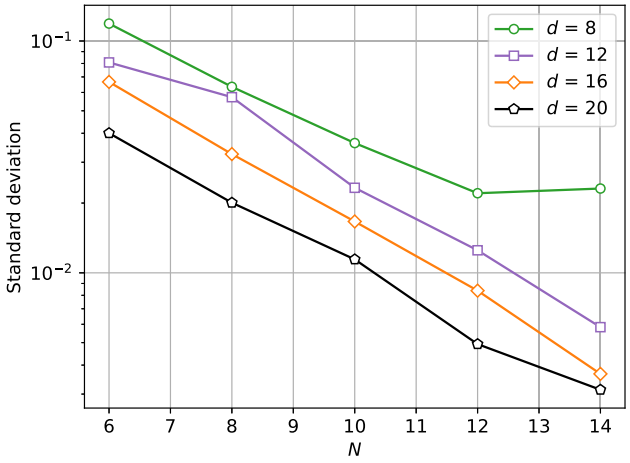


FIG. 20. Standard deviation of $F_{\mathrm{XEB}}^{\mathrm{lin}}$ computed over 100 random circuits and plotted versus system size for various depths. For this calculation, $F_{\mathrm{XEB}}^{\mathrm{lin}}$ is computed for each circuit by full density matrix simulation using a depolarizing noise model for the $2Q$ gates (with depolarizing parameter at each N rescaled to have the same error per layer as the measured value for H2 at $N = 56$). The standard deviation falls below 0.01 as N is increased for all but the shallowest ($d = 8$) circuits, which saturate at a slightly higher value with increasing N.

take to be a random variable that is Gaussian distributed about its experimental mean value $\varepsilon_{\mathrm{exp}} \approx 3.2 \times 10^{-3}$ with standard deviation $\mu \times \varepsilon_{\mathrm{exp}}$. Diagnostic and calibration data from the time period during which the data reported in Sec. IV were taken suggest a conservative upper bound $\mu \lesssim 0.2$. The set of all probability distributions over the collection of outcomes obtainable in this way is referred to as the *experimental model*.

We use this experimental model to perform a coverage analysis by simulating $N_E = 4000$ experiments, each with 50 circuits and 20 shots per circuit. For each dataset, we perform both aggregate bootstrapping and double bootstrapping using $r = 4000$ resamples. We then compute bootstrapped confidence intervals for each of the N_E simulated experiments from the synthetic bootstrap data and the empirical mean from that experiment and compute the fraction of experiments for which the true value (estimated by the mean across all experiments) is contained within that confidence interval, referring to that fraction as the *coverage*. A proper 1σ confidence interval must provide coverage $\geq 68.27\%$ for all distributions in the experimental model, and this level is denoted by the black dashed line in Fig. 21. In general, for the expected $\lesssim 20\%$ fluctuations in ε ($\mu \lesssim 0.2$), we see that aggregate resampling provides a confidence interval for $F_{\mathrm{XEB}}^{\mathrm{lin}}$ with the required coverage, while double resampling provides excessive coverage (resulting in an overly conservative confidence interval). For F_{MB}, the coverage of aggregate resampling falls off faster, requiring $\mu \lesssim 0.1$ to be adequate. In either case, for extremely large fluctuations in ε ($\mu = 1$), we see that aggregate resampling fails to provide adequate coverage

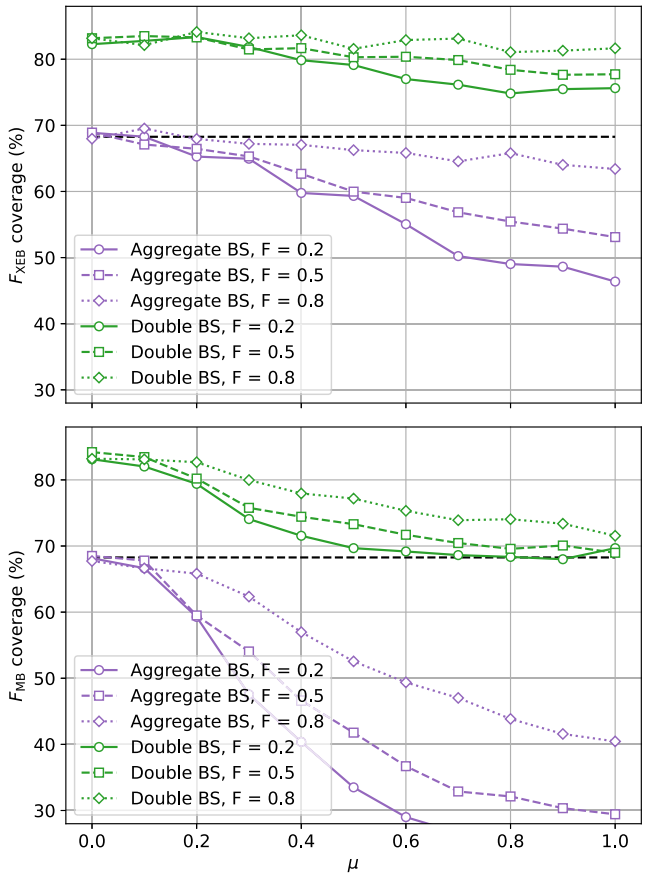


FIG. 21. Coverage analysis for aggregate bootstrapping and double bootstrapping applied to simulated experiments for both $F_{\mathrm{XEB}}^{\mathrm{lin}}$ (upper) and F_{MB} (lower). Double bootstrapping generally overcovers the data, most severely in the small μ limit where circuit-to-circuit fluctuations in the underlying distribution are small. Aggregate resampling generally provides an appropriate confidence interval for $\mu < 0.2$ for $F_{\mathrm{XEB}}^{\mathrm{lin}}$ (and $\mu < 0.1$ for F_{MB}) but can undercover the data in either case at large μ (especially for small values of the circuit fidelity F.).

(and, thus, underestimates the size of the correct 1σ confidence interval), while the coverage of double resampling remains adequate. Since we expect the actual circuit fluctuations in ε to be $\lesssim 20\%$, we believe that aggregate resampling is likely more appropriate for assigning confidence intervals to our $F_{\mathrm{XEB}}^{\mathrm{lin}}$ data, though it may slightly underestimate the appropriate confidence intervals for F_{MB}. Given that the confidence intervals are larger for $F_{\mathrm{XEB}}^{\mathrm{lin}}$, and in the interest of consistency, we opted to use the aggregate resampling method to report *all* 1σ error bars on both F_{MB} and $F_{\mathrm{XEB}}^{\mathrm{lin}}$ in the manuscript.

The larger variance estimate provided by the double resampling method can be explained analytically as follows. Consider the probability of any particular individual shot appearing k times when N_s shots are resampled. In the aggregate resampling method, this probability is binomially distributed:

$$\mathbb{P}_{\mathrm{agg}}(k) = \left(\frac{1}{N_s}\right)^k \left(1-\frac{1}{N_s}\right)^{N_s-k} \binom{N_s}{k}. \quad \text{(C2)}$$

In the double resampling method, the probability that a particular labeled shot appears k times depends on how many times its corresponding job is resampled. If there are N_j total jobs, with N_{per} shots per job and $N_s = N_j N_{\mathrm{per}}$,

$$\mathbb{P}_{\mathrm{doub}}(k) = \sum_{j=0}^{N_j} \left(\frac{1}{N_j}\right)^j \left(1-\frac{1}{N_j}\right)^{N_j-j} \binom{N_j}{j} \mathbb{P}(k|j), \quad \text{(C3)}$$

where $\mathbb{P}(k|j)$ is the probability that a given labeled shot appears k times given that its corresponding job is resampled j times. To compute $\mathbb{P}(k|j)$, one must consider all of the ways that the k appearances can be spread over the j job resamples. Note that the k appearances may be restricted to only a subset of the j resamples of size n_b "bins" as long as $k \le n_b N_{\mathrm{per}}$. Therefore, we compute $\mathbb{P}(k|j)$ by summing over the set of partitions of k of fixed size n_b, $P_{k,n_b} = \{(p_1, \ldots, p_{n_b}) \in (\mathbb{Z}^+)^{n_b} | \sum_i p_i = k\}$ as follows:

$$\mathbb{P}(k|j) = \sum_{n_b=0}^{j} \left[\binom{j}{n}_b \left(1-\frac{1}{N_{\mathrm{per}}}\right)^{(j-n_b)N_{\mathrm{per}}} \sum_{P_{k,n_b}} \mathbb{P}(k|P_{k,n_b}) \right]. \quad \text{(C4)}$$

The first term in the exterior sum counts the number of ways of choosing the n_b bins from the j job resamples. The second term is the standard binomial term accounting for the remaining $j - n_b$ jobs, each with N_{per} shots, where the labeled shot does not appear. The interior sum is over $\mathbb{P}(k|P_{k,n_b})$, the probability of the k appearances being distributed among bins according to each possible partition P_{k,n_b}. Note that the set of all such partitions includes "equivalent" partitions where the same set of p_i are permuted over the different bins n_b, which are counted separately in the sum. This final conditional probability is simply the product of binomial distributions over the n_b bins:

$$\mathbb{P}(k|P_{k,n_b}) = \prod_{i=1}^{n_b} \left(\frac{1}{N_{\mathrm{per}}}\right)^{p_i} \left(1-\frac{1}{N_{\mathrm{per}}}\right)^{N_{\mathrm{per}}-p_i} \binom{N_{\mathrm{per}}}{p}. \quad \text{(C5)}$$

The distributions $\mathbb{P}_{\mathrm{agg}}(k)$ and $\mathbb{P}_{\mathrm{doub}}(k)$ are depicted in Fig. 22 along with data averaged across 100 instances of numerical resampling experiments in which 1000 resamples were drawn from 1000 shots by either method and different job sizes. For all job sizes except $N_j = 1$ and $N_j = 1000$, where the double resampling method reduces to aggregate resampling, $\mathbb{P}_{\mathrm{doub}}(k)$ contains significantly higher weight at $k = 0$ and $k \ge 3$. One interpretation of this result is that, in a given resampling experiment using the double resampling method, the data tend to "clump," favoring multiple samples of the same shots and excluding other shots entirely, owing to the fact that each resampling experiment has a chance to exclude some jobs entirely. The higher variance estimate provided by the double resampling method can consequently can be viewed as an instance of Sheppard's correction [59] in which binned data tend to exhibit slightly higher variance estimates; a tendency to replace individual shots by multiple occurrences of the same shot is similar to replacing all data in a bin with the midpoint of the bin.

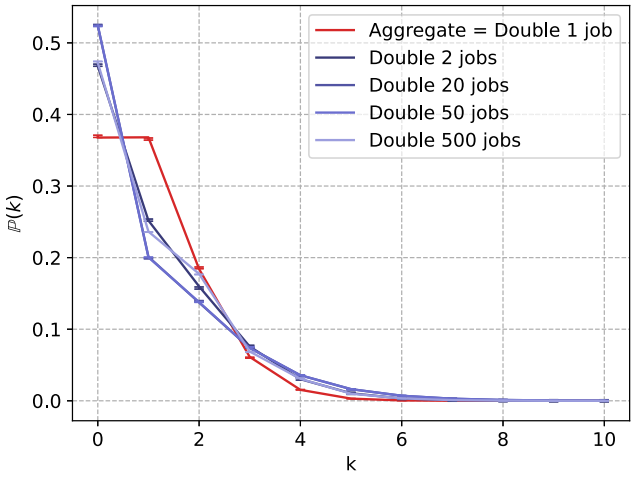


FIG. 22. Probability distributions $\mathbb{P}(k)$ of resampling a single labeled shot k times, depending on the aggregate or double resampling methods with 1000 shots binned into a variable number N_j of jobs and $1000/N_j$ shots per each job. In all cases except the $N_j = 1$ and $N_j = 1000$ limits, where the aggregate and double resampling methods are equivalent, the double resampling distributions contain significantly higher weight at $k = 0$ and $k \ge 3$. Numerically modeled points with error bars are overlaid on the solid line representing the analytic expressions in Eqs. (C2) and (C3).

APPENDIX D: CONSTRUCTION OF CIRCUITS WITH RANDOM GEOMETRIES

Considering each qubit as a node in a graph, a set of edges $\mathcal{E}$ connecting those nodes is said to be mutually disjoint if no two edges in $\mathcal{E}$ contain the same node. If we associate a $2Q$ gate to each edge in $\mathcal{E}$, those gates have mutually nonoverlapping support and can all be executed (in principle) in parallel. In a slight abuse of notation, we use $\mathcal{E}$ to also denote the set of $2Q$ gates associated to the edges in $\mathcal{E}$, and we refer to $\mathcal{E}$ as a single $2Q$ layer of a circuit. Each gate in $\mathcal{E}$ will be the H2 quantum computer's native perfect entangler $U_{ZZ}(\pi/2) = e^{-i(\pi/4)Z\otimes Z}$ (see Appendix G for further discussion regarding this gate choice). Similarly, let $\mathcal{S}$ denote a set of $1Q$ gates assigned to each qubit; we refer to $\mathcal{S}$ as a single $1Q$ layer of a circuit.

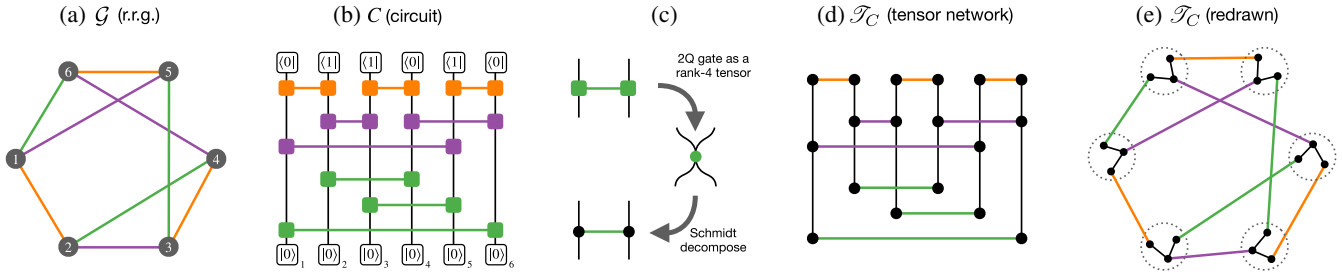


FIG. 23. (a) To construct a depth-3 circuit on six qubits, we first select a random 3-regular graph on n vertices, denoted $\mathcal{G}$, and assign to it a proper edge coloring using three colors. (b) $\mathcal{G}$ induces the circuit C shown by associating each vertex (edge) in $\mathcal{G}$ to a qubit ($2Q$ gate) in C; each $2Q$-gate layer of C is in correspondence with a given color in the coloring of $\mathcal{G}$. Qubits are assumed (without loss of generality since random $1Q$ gates are applied immediately after initialization) to be initialized in the $|0\rangle$ state, and layers of $1Q$ gates in C have been absorbed into the $2Q$ gates for clarity. The probability that the circuit outputs a given product state (here, the state $|011010\rangle$) can be represented as a tensor network as shown in (c) by associating each $2Q$ gate to a rank-4 tensor. (d) Upon Schmidt decomposing those gates and absorbing the rank-1 tensors corresponding to input and output states into adjacent $2Q$ gate tensors, we obtain the tensor network $\mathcal{T}_C$. (e) An equivalent redrawing of $\mathcal{T}_C$ reveals its similarity to the underlying graph $\mathcal{G}$ from which the circuit was generated.

Each $1Q$ gate $S \in \mathcal{S}$ is chosen to be a Haar-random SU(2) gate, which can be decomposed as $S = R_z(\psi)\mathrm{U1q}(\theta,\phi)$. Here,

$$\mathrm{U1q}(\theta,\phi) = e^{-i(\theta/2)(\hat{X}\cos\phi+\hat{Y}\sin\phi)} \tag{D1}$$

is our native $1Q$ gate, and $R_z(\psi)$ is a z rotation that is performed in software (i.e., by pushing the z rotation through future $1Q$ gates and modifying their phases accordingly). A random depth-d circuit is then defined by interleaving such layers as

$$C = \mathcal{S}_d\mathcal{E}_d\mathcal{S}_{d-1}\mathcal{E}_{d-1}\ldots\mathcal{S}_1\mathcal{E}_1\mathcal{S}_0. \tag{D2}$$

A powerful feature of QCCD architecture is the flexibility to choose which pairs of qubits are gated in a given $2Q$ layer: Regardless of the physical locations of the qubits after the previous layer, any set $\mathcal{E}$ can contain arbitrary pairs and the associated gates will be executed directly (e.g., without logical SWAP gates) by physically rearranging the ions such that the qubits associated to each $2Q$ gate in $\mathcal{E}$ are located next to each other. In this work, we consider the ensemble of random circuits in which the sets $\mathcal{E}_1,\ldots,\mathcal{E}_d$ are obtained via the following procedure:

(1) Select a random d-regular graph $\mathcal{G}$ on N vertices.
(2) Find a proper edge coloring of $\mathcal{G}$ with d colors.
(3) Assign a unique index $j = 1,\ldots,d$ to each color in the edge coloring, and put each edge with color j into the set $\mathcal{E}_j$.

A proper edge coloring is defined as an assignment of a color to each edge in a graph such that no two edges of the same color impinge on the same node, and, hence, each $\mathcal{E}_j$ is a proper (mutually disjoint) layer of $2Q$ gates [see Fig. 23(a) for an example of a proper edge coloring of a 3-regular graph on six nodes using three colors]. By Vizing's theorem, $\mathcal{G}$ is guaranteed to have a proper edge coloring using a minimum number of colors equal to either d (in which case $\mathcal{G}$ is referred to as class I) or $d+1$ (in which case $\mathcal{G}$ is referred to as class II). Note that in step 2 of the above procedure we implicitly assume that the sampled graph is class I. While this assumption is not guaranteed to be true, it is known that random regular graphs of degree $d \geq 3$ are asymptotically almost always (i.e., with probability approaching 1 for $N \to \infty$) class I [60,61], and all d-regular graphs with even N and $d \geq 2\lceil N/4\rceil - 1$ are provably class I [62]. Empirically, we are able to find proper d colorings of all but a very small subset of randomly sampled graphs in the ranges of (N,d) that we have considered. In the rare event that no proper d coloring is found in step 2, we reject the graph and return to step 1.

APPENDIX E: BOUNDS ON CONTRACTION DIFFICULTY OF RANDOM-GEOMETRY CIRCUITS

To any unitary quantum circuit C, one can associate a tensor network $\mathcal{T}_C$ in which gates are complex-valued tensors and wires (qubits) are the indices or legs of those tensors. In this representation, $1Q$ gates become rank-2 tensors (one incoming and one outgoing qubit wire) and $2Q$ gates become rank-4 tensors (two incoming and two outgoing qubit wires). The initial state $|\psi_j\rangle$ of each input qubit j is a rank-1 tensor, and the output of the tensor network (uncontracted legs) represents the quantum state at the output of the circuit, $|\Psi\rangle = C\otimes_j|\psi_j\rangle$. The overlap between $|\Psi\rangle$ and a particular tensor product of single-qubit states $\otimes|\varphi_j\rangle$ is then obtained by contracting each output leg of the tensor network with a rank-1 tensor corresponding to the $1Q$ state $|\varphi_j\rangle$ associated to that qubit. As an example, in Fig. 23(b), we show the depth $d=3$ quantum circuit C induced by the 3-regular graph $\mathcal{G}$ shown in Fig. 23(a), as described in Appendix D. The tensor network associated to this circuit, denoted $\mathcal{T}_C$, is shown in Fig. 23(d). Note that we have decomposed each rank-4 gate tensor into a tensor

product of two rank-3 tensors as shown in Fig. 23(c). For the $U_{ZZ}(\pi/2)$ gates used here, this decomposition has Schmidt rank two. Therefore, both the colored lines (originating from gates) and the black lines (originating from qubit wires) have dimension 2, and we need not distinguish between them when determining the time or memory cost of tensor contraction. We have also absorbed the rank-1 input and output tensors into the nearest gate tensor. Figure 23(e) shows how the tensor network $\mathscr{T}_C$ is structurally similar to the graph $\mathcal{G}$ from which the circuit C was generated.

1. Lower bound

To arrive at the TN $\mathscr{T}_C$ in the form shown in Fig. 23(e), we have already decomposed each rank-4 $2Q$-gate tensor into two rank-3 tensors. However, we could also consider the contraction cost of the original circuit TN ($\mathscr{T}_C^0$, not shown in Fig. 23). The TN $\mathscr{T}_C$ cannot be harder to contract than $\mathscr{T}_C^0$, because one viable contraction path for $\mathscr{T}_C$ consists of first grouping each gate back to a single rank-4 tensor, thereby achieving $\mathscr{T}_C^0$ as an intermediate tensor (and this process occurs at much lower cost than future operations in contracting $\mathscr{T}_C^0$). Therefore, the contraction cost of $\mathscr{T}_C$ lower bounds the contraction cost of the original circuit TN $\mathscr{T}_C^0$, and we need only to lower bound the cost of contracting $\mathscr{T}_C$ in the form shown in Fig. 23(e).

Below, we prove that, for $d \geq 3$ and with probability (over the distribution of circuits considered) approaching 1 as $N \to \infty$, the contraction cost for our circuits obeys

$$\mathcal{C}_d = \lim_{N\to\infty} \frac{\log_2(\text{cost})}{N} \geq c \tag{E1}$$

for some constant $0 < c < 1$. Note that, for simplicity, the definition of $\mathscr{C}_d$ used here (and in the following section) differs slightly from that in the manuscript in that we do not normalize the cost by the number of gates. However, such a normalization generates only an additive $\sim \log(N)/N$ correction to the lhs, and, therefore, Eq. (E1) will hold for $\mathscr{C}_d$ as defined in the manuscript whenever it holds here. This result should be compared to the observation that $\mathscr{C}_d = 0$ in *any* local and finite-dimensional geometry. The underlying methodology is related to the work of Markov and Shi (see Theorem 1.7 in Ref. [27])—however, the proof is more direct and requires very little graph-theoretic machinery; ultimately, it is based on the recognition that for circuits with gates drawn from graphs with good expansion properties, TN contraction requires forming intermediate tensors that either cut an extensive number of qubit wires, gates, or both.

$\mathscr{T}_C$ is composed of Nd original tensors, and its contraction into a scalar (a single rank-0 tensor whose value is the amplitude $\langle\Phi|C|\Psi\rangle$ for a particular output state $|\Phi\rangle$) proceeds as a sequence of individual contractions that can be represented as a rooted binary tree (see Fig. 24 for an example), with the original tensors as leaves and the final scalar as the root. Each vertex in the tree corresponds to a merged tensor during the contraction process, and its children (the two tensors merged to make it) are connected to it by edges. Each vertex can be labeled by its descendants, e.g., which of the leaves (original tensors) have been absorbed into the associated tensor, and those leaves are shown explicitly inside the vertices in Fig. 24. The tensor obtained by the jth contraction is denoted $\mathscr{T}_j$, and we define its size $|\mathscr{T}_j|$ as its number of descendants (i.e., the number of original tensors that merged to form it). We refer to the number of legs of the tensor $\mathscr{T}_j$ as its *rank* r_j. Because all edges of $\mathscr{T}_C$ have dimension 2 (due to our gates being rank-2 entanglers), the contraction cost (measured in FLOPs) associated to a particular contraction tree is lower bounded as

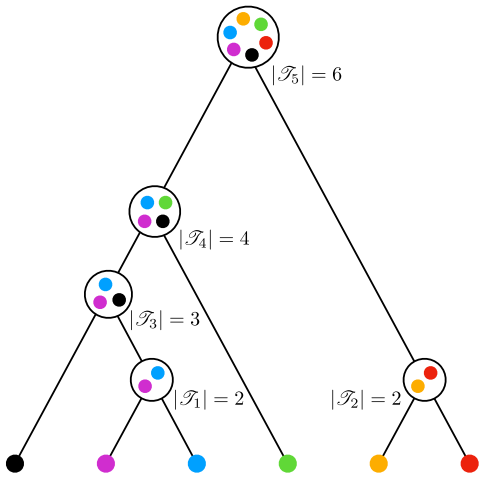


FIG. 24. Representation of the contraction of a tensor network $\mathscr{T}$ into a scalar as a rooted binary tree flowing from bottom (leaves) to top (root) (here, we assume that $\mathscr{T}$ has no open legs so that it can be contracted to a scalar, as in the case of a TN representing the probability amplitude of a given state at the output of a quantum circuit). Here, $\mathscr{T}$ is chosen to have six tensors, which are represented by the leaves of the tree (each is given a unique color), while every other vertex represents a merged tensor formed by contracting any legs shared by its two children. The root represents the final (scalar) value of the contraction.

$$\text{cost} \geq \max_j 2^{r_j}. \tag{E2}$$

The cost of contracting $\mathscr{T}_C$ is obtained by minimizing the contraction cost over all possible contraction trees. Note that there must be a smallest value of j such that $|\mathscr{T}_j| \geq Nd/3$ (i.e., as one proceeds with the contraction, there must be a first time when two merged tensors result in a final tensor of size at least $Nd/3$), and we call this intermediate tensor $\mathscr{T}_{\min}$. Since every other tensor has (by assumption) size less than $Nd/3$ when the jth contraction happens, we also know that $|\mathscr{T}_{\min}| \leq 2Nd/3$. Our goal here is to ensure

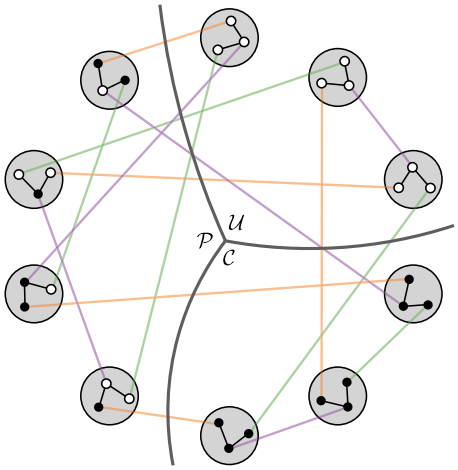


FIG. 25. Example tensor network for a depth-3 random circuit on ten qubits, with a possible intermediate tensor in the contraction process denoted as the collection of black tensors (e.g., they will all be contracted together into a single tensor during the contraction process). There is always an intermediate tensor built by contracting at least $1/3$ and at most $2/3$ of the original tensors, and in this example we are depicting an intermediate tensor built by contracting exactly half ($Nd/2 = 15$) the original tensors. Each gray disk, which we refer to as a bag, is associated with a particular qubit and contains all tensors having legs originating from that qubit's wire. Those bags can be subdivided into sets $\mathcal{C}$, $\mathcal{U}$, and $\mathcal{P}$, according to whether they are fully contracted, completely uncontracted, or partially contracted, respectively.

that both the set of original tensors contained in $\mathscr{T}_{\min}$ and the set of all other tensors are *both* extensive, which will ultimately be used to ensure that an associated subset of the vertices of $\mathcal{G}$ has a large edge boundary (which requires that the subset and its complement are both large). We could have used any number of tensors $k \times Nd$ with $0 < k < 1/2$ (rather than our choice $k = 1/3$) when defining $\mathscr{T}_{\min}$ ($1/2$ is problematic, since then $\mathscr{T}_{\min}$ could contain almost every original tensor).

For simplicity of illustration, let us assume that $|\mathscr{T}_{\min}| = (Nd/2)$; i.e., $\mathscr{T}_{\min}$ is composed of exactly half of the primitive tensors. Figure 25 shows an example of $\mathscr{T}_C$ for a ten-qubit depth-3 circuit in which we have labeled the $Nd/2$ original tensors forming $\mathscr{T}_{\min}$ as filled circles (•) and all of the original tensors not merged into $\mathscr{T}_{\min}$ as open circles (◦). To each qubit in the circuit, we can associate a *bag* of tensors belonging to the gates that act on that qubit; in the depth-3 example in Fig. 25, each bag (gray disk) contains exactly three tensors. The set of all bags can be decomposed as $\mathcal{C} \cup \mathcal{U} \cup \mathcal{P}$, such that $\mathcal{C}$ contains the bags for which all constituent tensors have been contracted into $\mathscr{T}_{\min}$, $\mathcal{U}$ contains the bags for which none of the constituent tensors have been contracted into $\mathscr{T}_{\min}$, and $\mathcal{P}$ contains the bags for which some but not all constituent tensors have been contracted into $\mathscr{T}_{\min}$.

It is helpful to consider the strategy of state vector simulation, in which all gates of a given color are absorbed sequentially, causing all bags to be in $\mathcal{P}$. More generally, each bag of type $\mathcal{P}$ is associated with at least one qubit wire protruding from the intermediate tensor $\mathscr{T}_{\min}$, and choosing the set $\mathcal{P}$ to be large corresponds to a predominantly spacelike slicing of the network (e.g., created by slicing in a way that divides tensors at later times from those at earlier times), suffering a cost that is exponential in the number of qubits. A productive contraction strategy for sufficiently shallow spatially local lattices is to avoid bags of type $\mathcal{P}$ entirely, corresponding to making timelike cuts through the tensor network such that only gate wires protrude from $\mathscr{T}_{\min}$. Judiciously choosing the set C (to have a small edge boundary) can greatly reduce the contraction cost relative to the state vector strategy. The fundamental reason why Eq. (E1) holds for RG circuits is that *neither* strategy (nor any other strategy, as we show below) succeeds in producing an intermediate tensor $\mathscr{T}_{\min}$ with a number of legs that is sublinear in N. The timelike strategy can be seen to fail by considering the extreme limit in which $\mathcal{P}$ is empty: The number of gate wires exiting $\mathscr{T}_{\min}$ is then determined by the number of edges connecting bags in $\mathcal{U}$ to bags in $\mathcal{C}$, and this number is guaranteed to scale linearly with the size of $\mathcal{C}$ whenever the underlying graph $\mathcal{G}$ is a good expander, as is the case for random regular graphs. The above reasoning is somewhat heuristic, but it can readily be turned into a proof of Eq. (E1). Specifically, we need to show that, with probability (over the considered class of graphs considered) approaching 1 for large N, no contraction strategy can yield $\mathscr{T}_{\min}$ with rank $r_{\min} < cN$ for some constant c.

If we consider $\mathscr{T}_C$ as a graph (tensors → nodes and legs → edges) and denote the set of black nodes in Fig. 25 as V, the rank of $\mathscr{T}_{\min}$ is the size of the edge boundary of V, denoted by $|\partial V|$, and Eq. (E1) will be true as long as $|\partial V| \geq cN$. The edge boundary ∂V contains all colored edges that originate from $\mathcal{C}$ and terminate on vertices in $\mathcal{U}$ or unfilled vertices in $\mathcal{P}$. The number of such edges is lower bounded by $\max(0, |\partial\mathcal{C}| - d|\mathcal{P}|)$. Since every bag in $\mathscr{P}$ also contributes at least one black edge to ∂V (not already counted in the colored edges considered above), we conclude that

$$|\partial V| \geq |\mathcal{P}| + \max(0, |\partial\mathcal{C}| - d|\mathcal{P}|). \tag{E3}$$

The first and second terms in Eq. (E3) capture the costs of attempting to contract the tensor network in a timelike (Schrödinger) or spacelike direction, respectively. The edge boundary of $\mathcal{C}$ is constrained relative to its size by the edge isoperimetric number of $\mathcal{G}$, defined as

$$I(\mathcal{G}) = \min_U |\partial U|/|U|, \tag{E4}$$

with U ranging over all subsets of nodes in $\mathcal{G}$ containing at most half of the vertices in $\mathcal{G}$. Either $\mathcal{C}$ or its complement satisfies the constraint of having at most half of the vertices in $\mathcal{G}$. The complement of $\mathcal{C}$ is lower bounded in size by $N/3$ (because $|V| < 2Nd/3$), and, therefore,

$$|\partial C| \geq I(\mathcal{G})\ \min(|\mathcal{C}|, N/3). \tag{E5}$$

The notion that random regular graphs are (with high probability) good expanders can be formalized in the following theorem by Bollobás [63] regarding the isoperimetric number of $\mathcal{G}$.

Good edge expansion of random d-regular graphs. Consider the distribution of d-regular graphs on N nodes sampled uniformly at random, and define $\eta(d) = 2\sqrt{(\log 2)/d}$. For all $d \geq 3$, and with probability approaching 1 as $N \to \infty$, the isoperimetric number $I(\mathcal{G})$ of a graph $\mathcal{G}$ sampled from this distribution is bounded below as

$$I(\mathcal{G}) \geq \frac{d}{2}(1 - \eta(d)). \tag{E6}$$

Combining Eq. (E6) with Eqs. (E3) and (E5) yields the lower bound

$$|\partial V| \geq |\mathcal{P}| + d \times \max\left(0, \frac{\min(|\mathcal{C}|, N/3)[1 - \eta(d)]}{2} - |\mathcal{P}|\right). \tag{E7}$$

Recall that, by the definition of $\mathcal{T}_{\min}$, the size of V is constrained to satisfy

$$\frac{dN}{3} \leq |V| \leq \frac{2dN}{3}. \tag{E8}$$

Since each bag in $\mathcal{C}$ contributes exactly d vertices to V while each bag in $\mathcal{P}$ contributes at most $d - 1$, we conclude that

$$|\mathcal{C}| \geq \frac{N}{3} - |\mathcal{P}|, \tag{E9}$$

from which we can further bound $|\partial V|$ as

$$|\partial V| \geq |\mathcal{P}| + d \times \max\left(0, \frac{[N/3 - |\mathcal{P}|][1 - \eta(d)]}{2} - |\mathcal{P}|\right). \tag{E10}$$

Upon varying $|\mathcal{P}|$, the rhs is minimized (for $d \geq 3$) when the second argument of the max() function vanishes, ultimately giving

$$|\partial V| \geq \frac{N[1 - \eta(d)]}{9}. \tag{E11}$$

Note that we have not attempted to make the bound even reasonably tight, and we expect it is possible to derive a bound in which the rhs tends to N as $d \to \infty$.

2. Upper bound

The previous section demonstrates that the asymptotic contraction complexity density $\mathscr{C}_d$ is finite but does not address how the density approaches the worst-case upper bound from state vector simulation ($\mathscr{C}_d \to 1$) as the depth increases. Figure 26 shows numerical evidence that the complexity density saturates and also that the saturated value quickly approaches 1 as d increases. It is natural to ask precisely *how* $\mathscr{C}_d \to 1$ as $d \to \infty$: Does $1 - \mathscr{C}_d$ vanish smoothly as $d \to \infty$, and if so, does it decay polynomially or exponentially? Or does it vanish abruptly at some critical value $d = d_c$?

Here, we show that $1 - \mathscr{C}_d$ cannot vanish abruptly by proving that individual circuit instances obey the upper bound

$$\mathscr{C}_d \leq 1 - 1/2^d. \tag{E12}$$

This bound leaves open the possibility that $1 - \mathscr{C}_d$ vanishes exponentially as $d \to \infty$; we find this scenario likely but have no proof of it. The proof of Eq. (E12) is by analysis of an explicit contraction ordering inspired by the qubit reuse algorithm described in Ref. [64].

Pick any pair of qubits j_1, k_1 that are gated together in the final layer of gates and contract all tensors corresponding to gates in those qubits' past causal cone. A representative example of such a causal cone is shown for a ten-qubit depth-3 circuit in Fig. 27. Denote the set of output qubits by $\mathcal{O}_1 = \{j_1, k_1\}$ and the collection of input qubits in the past causal cone of $\mathcal{O}_1$ by $\mathcal{I}_1$. It is easy to see that these sets satisfy

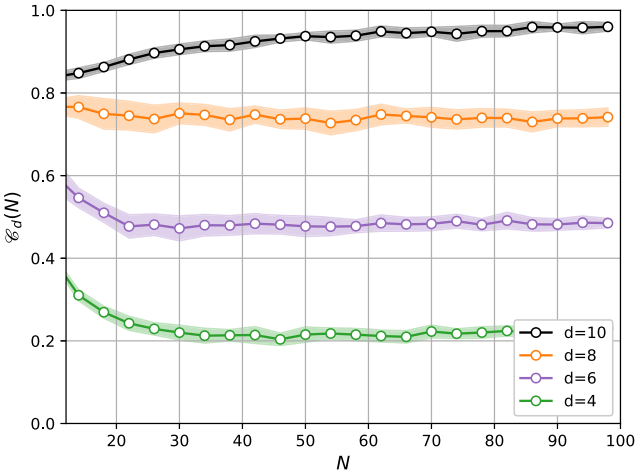


FIG. 26. Complexity density as a function of N for circuit depths $d = 4, 6, 8, 10$, showing saturation to a constant as $N \to \infty$. Each point is the mean complexity density over 20 random circuits (shaded regions represent one standard deviation).

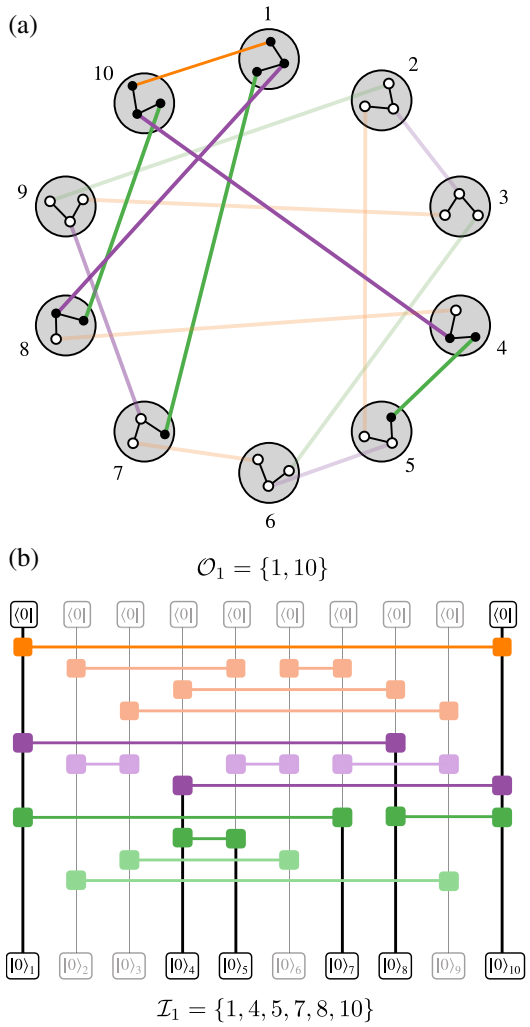


FIG. 27. Light-cone-based contraction ordering yielding the upper bound in Eq. (E12).

$$|\mathcal{O}_1| = 2, \tag{E13}$$

$$|\mathcal{I}_1| \leq \min(2^d, N). \tag{E14}$$

This contraction can be performed in a timelike direction with at most $|\mathcal{I}_1|$ legs protruding from any intermediate tensor (corresponding to unterminated wires of the input qubits in $\mathcal{I}_1$). Note that the final tensor formed this way has at most $|\mathcal{I}_1| - 2$ open legs, since we close the two outputs associated with qubit j_1 and k_1. Now we pick a new pair of output qubits $j_2, k_2 \notin \mathcal{O}_1$ that are gated together in the final layer, define $\mathcal{O}_2 = \{j_2, k_2\} \cup \mathcal{O}_1$, and define $\mathcal{I}_2$ to be the set of all qubits in the past casual cone of $\mathcal{O}_2$. Since $\mathcal{I}_2 \backslash \mathcal{I}_1$ has at most 2^d qubits, continuing the contraction process in a timelike direction with all gates in $\mathcal{I}_2$ that have not already been contracted cannot add more than 2^d legs to the existing tensor. Therefore,

$$|\mathcal{O}_2| = 4, \tag{E15}$$

$$|\mathcal{I}_2| \leq \min(N, 2 \times 2^d), \tag{E16}$$

and no intermediate tensor can have more than $|\mathcal{I}_2| - |\mathcal{O}_1|$ legs (the "$-|\mathcal{O}_1|$" contribution because two indices corresponding to the outputs at qubits in $\mathcal{O}_1$ have been fixed). Iterating this process, we find that at stage ℓ

$$|\mathcal{O}_\ell| = 2 \times \ell, \tag{E17}$$

$$|\mathcal{I}_\ell| \leq \min(N, \ell \times 2^d), \tag{E18}$$

and no intermediate tensor can have had more than $|\mathcal{I}_\ell| - |\mathcal{O}_{\ell-1}|$ legs. At some point in this process, the size of $\mathcal{I}_\ell$ will be equal to N, but that cannot happen unless $\ell \geq N/2^d$. At this stage, the largest number of legs in any intermediate tensor is upper bounded by

$$|\mathcal{I}_\ell| - |\mathcal{O}_{\ell-1}| \leq N - 2(\ell - 1) \tag{E19}$$

$$\leq N - 2(N/2^d - 1) \tag{E20}$$

$$\leq N(1 - 1/2^d) + 2, \tag{E21}$$

giving

$$\mathscr{C}_{d,N} \leq 1 - 1/2^d + 2/N. \tag{E22}$$

Ignoring the latter term in the limit $N \to \infty$ yields Eq. (E12).

APPENDIX F: SIMPLE MODELS OF EXACT CONTRACTION COST

In Sec. VI of the manuscript, we report comparisons of achievable effective qubit numbers for 2D and RG quantum circuits. Those comparisons are based on a simplified model of memory-unconstrained contraction cost that we now describe. Figures 28(a) and 28(d) show exact contraction costs determined using cotengra for 2D circuits and RG circuits, respectively (as described in the main text).

In 2D, the complexity density $\mathscr{C}_{d,N}$ is expected to scale as $\min(1, c \times d/\sqrt{N})$, with c an *a priori* undetermined constant. Therefore, defining $\tilde{d} = d/\sqrt{N}$, we expect a plot of $f_N(\tilde{d}) \equiv \mathscr{C}_{\tilde{d}\sqrt{N},N}$ to show collapse to a universal curve (when plotted vs $\tilde{d}$) as N becomes large enough to suppress boundary effects. Figure 28(b) shows exactly this behavior, and based on the largest-N results we choose a (generous) envelope function $f(\tilde{d}) = 1.1 \min(1.0, 0.35\tilde{d})$ [black dashed line in Fig. 28(b)]. Assuming this heuristic model of the complexity density and converting back into $\log_{10}$ (contraction cost), we obtain the plot in Fig. 28(c), showing good qualitative agreement with the behavior in Fig. 28(a).

For random geometry circuits, the complexity density is expected to scale independently of N for large N, approaching 1 for large d with an *a priori* unknown functional form.

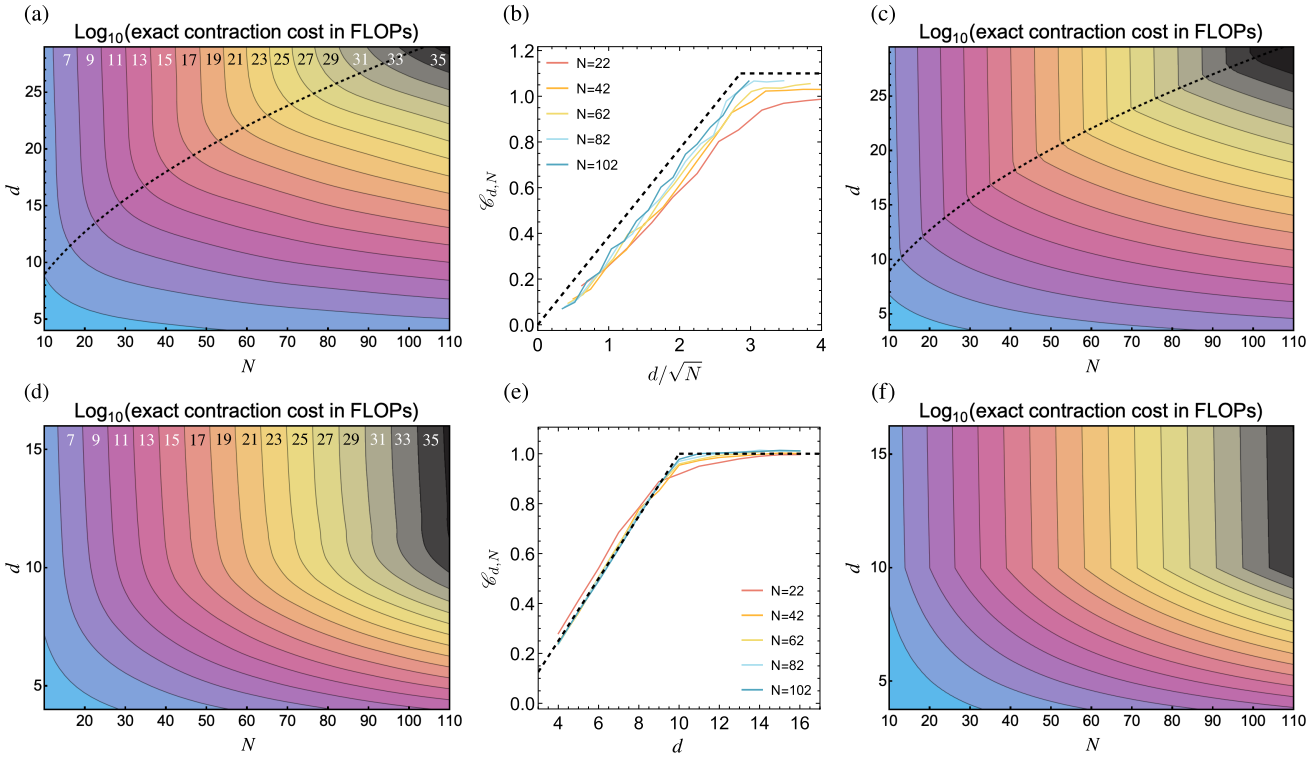


FIG. 28. Models of the exact contraction cost for both 2D (top) and RG (bottom) circuits. (a) and (d) show the exact contraction cost estimated using cotengra for 2D and RG circuits, respectively. (b) and (e) show slices from those plots at various N, showing collapse to the simple models described in the text at large N. (c) and (f) show plots of the cost inferred from those simple models with parameters fit from (b) and (e), showing good qualitative agreement with the results in (a) and (d).

Empirically, we see that $\mathscr{C}$ grows roughly linearly until $d \approx 10$, and after this depth it saturates extremely quickly to 1. Together with the observation that the cost is small and constant at $d = 1, 2$ [27], this behavior motivates us to model the cost with a simple functional form $\mathscr{C}_{d,N} \approx a \times \min[1, c(d-2)]$. Evaluated with the constants $(a, c) = (1, 0.125)$, this ansatz gives the black dashed curve in Fig. 28(e), which agrees well with the largest N calculations we performed. When converted into $\log_{10}$ (contraction cost), this ansatz yields the plot in Fig. 28(f), showing good qualitative agreement with the results obtained using cotengra in Fig. 28(d).

APPENDIX G: IMPACT OF GATE CHOICE ON SIMULATION HARDNESS

At sufficiently high depths, optimal TN contraction is expected to proceed in a timelike direction, exhibiting computational costs approaching state vector simulation. In this limit, gate choice will have minimal impact on the cost of exact contraction. On the other hand, at low depths the optimal contraction strategy will generally not be timelike, in which case it may benefit greatly from $2Q$ gates having low-rank decompositions. A simple example is the 1D circuit shown in Fig. 29(a), which always admits the sideways contraction ordering shown in Fig. 29(c) with cost 2^d. Decomposing the $2Q$ gates with Schmidt rank r_s enables a sideways contraction ordering with cost $\sim r_s^{d/2}$. For $2Q$ gates with $r_s = 4$, such as the iSWAP gate, $r_s^{d/2} = 2^d$ and the contraction cost is the same for Figs. 29(c) and 29(d). For $2Q$ gates with $r_s = 2$, such as the $U_{ZZ}(\pi/2)$ gate, $r_s^{d/2} = 2^{d/2}$. Therefore, in this simple 1D example, using a rank-4 entangling gate rather than a rank-2 entangling gate effectively doubles the circuit depth.

The situation for circuits with more complicated geometries is less clear, and we have resorted to numerically studying the contraction cost dependence on gate rank by using the heuristic contraction cost optimizers provided by cotengra. Figure 30 compares the exact contraction cost of RG quantum circuits built of $2Q$ gates having rank-2 (circles) and rank-4 (squares) decompositions. A natural choice of rank-4 gate would be the iSWAP gate, as it has a uniform (rank-4) Schmidt spectrum and, therefore, resists approximation strategies based on low-rank gate approximations. However, the iSWAP gate can be made rank-2 by factoring out a SWAP gate. Factoring SWAP out of every $2Q$ gate would effectively map the original circuit on one random geometry and with rank-4 gates onto a new circuit on a *different* but still random geometry with rank-2 gates, and this new circuit should have a similar exact contraction cost as the original circuit with rank-2 gates. To avoid this

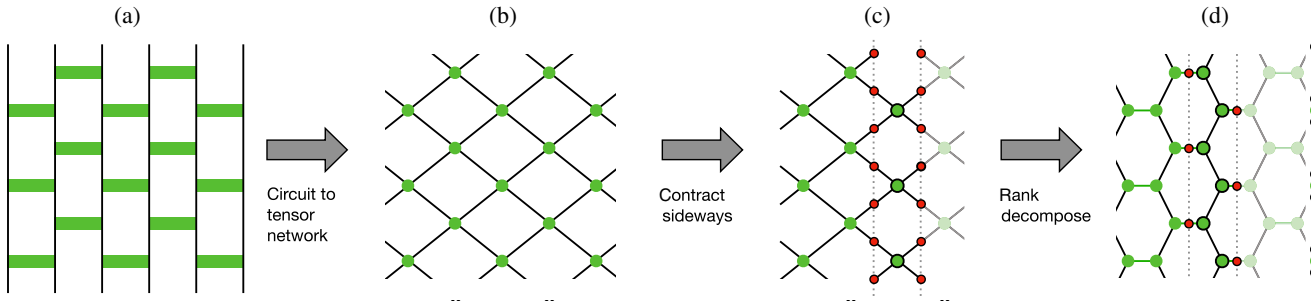


FIG. 29. Example of exact contraction cost dependence on gate rank for a 1D circuit. The circuit in (a) can be rewritten as the tensor network in (b) by assigning wires to legs and a rank-4 tensor to each $2Q$ gate. For sufficiently low depths, the tensor network in (b) can be contracted most efficiently by proceeding sideways. In (c), the dashed lines show intermediate tensors before and after application of tensors in a single vertical slice (black outlined), and red circles show open indices both before and after application of all tensors in that slice. Each open index takes values in $\{0, 1\}$, and there are $\sim d$ such indices for a depth-d circuit, giving an overall contraction cost $\sim 2^d$. The rank-4 gate tensors can always be Schmidt decomposed into two rank-3 tensors, with those two tensors sharing a leg taking values $\{1, \ldots, r_s\}$, with r_s the Schmidt rank of the gate, as shown in (d). Sideways contraction now leads to intermediate tensors with $\sim d/2$ open indices, for a total contraction cost scaling as $\sim r_s^{d/2} \leq 2^d$.

simplification strategy, we must use a rank-4 gate that remains rank-4 under SWAP factorization, and we choose a partial iSWAP for the numerical study in Fig. 30. This choice is essentially arbitrary: in the space of all possible $2Q$ gates, all but a set of measure zero has the aforementioned property (being rank 4 and remaining rank 4 under SWAP factorization), and so nearly any $2Q$ gate would lead to the same exact contraction costs given by the black squares in Fig. 30.

From Fig. 30, we see that, all else being equal, rank-4 gates should be preferred as they lead to higher contraction costs at fixed circuit depth. But all else is not equal; Quantinuum's native $U_{ZZ}(\theta)$ gate is not rank 4, and building a rank-4 gate requires at least two native gates.

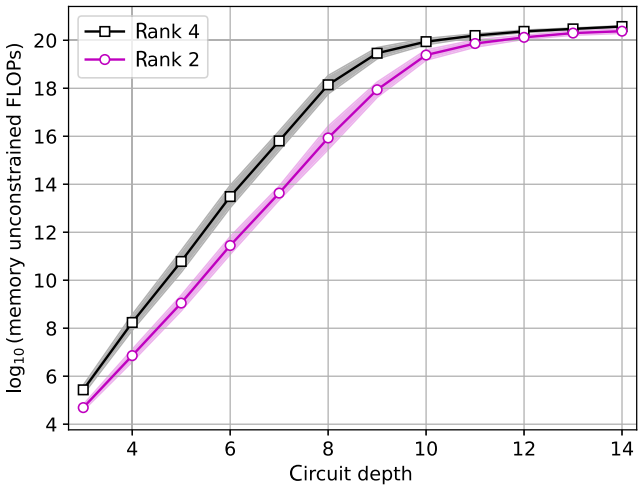


FIG. 30. Comparison of memory-unconstrained exact contraction costs for RG quantum circuits on 56 qubits with rank-2 and rank-4 gates. Shaded regions represent the standard deviation on $\log_{10}$ (FLOPs) computed across 20 random circuit realizations at each depth.

In principle, it should be sufficient to append a single relatively weak entangler in a different basis to our native gate [e.g., $U_{ZZ}(\theta) \to U_{ZZ}(\theta)U_{XX}(\delta)$]. However, even in the limit $\delta \to 0$ our partial entangler has a nonzero average infidelity of $\epsilon_0 \sim 4.6(6) \times 10^{-4}$ [8]. Thus, one can always achieve the same circuit fidelity using just the native $2Q$ gate at a depth increased by a factor of $r \equiv [\varepsilon_{2Q} + (5/4)\epsilon_0]/\varepsilon_{2Q}$. Rescaling the depths of the rank-4 (black) curve in Fig. 30 by a factor of r yields a result very similar to the rank-2 curve, suggesting that the advantage (in depth reduction at fixed contraction cost) of using a rank-4 gate is largely offset by the disadvantage (in achievable depth at fixed fidelity) due to the relatively lower rank-4 gate fidelity. Therefore, we have opted to take the simpler route of applying a single $U_{ZZ}(\pi/2)$ to each qubit pair gated in each layer of our circuits (as described in Appendix D).

APPENDIX H: COMPARISON OF VARIOUS CONTRACTION-ORDER OPTIMIZERS

Finding truly optimal contraction orders for a generic tensor network is a hard problem, and the cost estimates for exact contraction given in the main manuscript are all based on heuristic algorithms deployed by cotengra for contraction order optimization. To understand how the resulting FLOP costs are situated relative to other state-of-the-art TN contraction optimization packages, we have made a variety of direct performance comparisons between cotengra, cuQuantum [65], tamaki [66], and multilevel [67]. State-of-the-art tensor network contraction algorithms used for simulating random circuit sampling use various techniques to achieve sublinear costs in the number of samples [29,68–72], including sparse states, caching, etc. Here, we focus on single amplitude contraction optimization, since very few samples (20 in this paper) are produced from each unique circuit.

The most relevant cost comparison is actual contraction time, which requires executing the contraction order obtained on classical computing hardware. However, optimizers such as tamaki and multilevel do not integrate tensor network slicing directly, which means that the contraction order is not readily usable. Another metric that can be directly compared across all four optimizers considered is the minimum width of a tree decomposition obtained by the optimizer, which determines the largest intermediate tensor size required for the contraction order found.

We choose a randomly generated 56 qubit depth-10 RG circuit and allow the corresponding tensor network (without simplification, since tamaki and multilevel do not support it yet) to be optimized by each optimizer. Figure 31(a) shows the width of the lowest-width tree decomposition obtained after a given amount of optimization time and demonstrates that cuQuantum [65] and cotengra [31,73] generally outperform tamaki and multilevel given fixed optimization time. Further, these optimizers supports slicing and tensor network simplification out of the box. However, we have not been able to successfully utilize simplification of circuits containing many diagonal gates (such as U_{ZZ} gates) in cuQuantum. Therefore, we concentrate on comparing cuQuantum against cotengra for tensor networks without simplification.

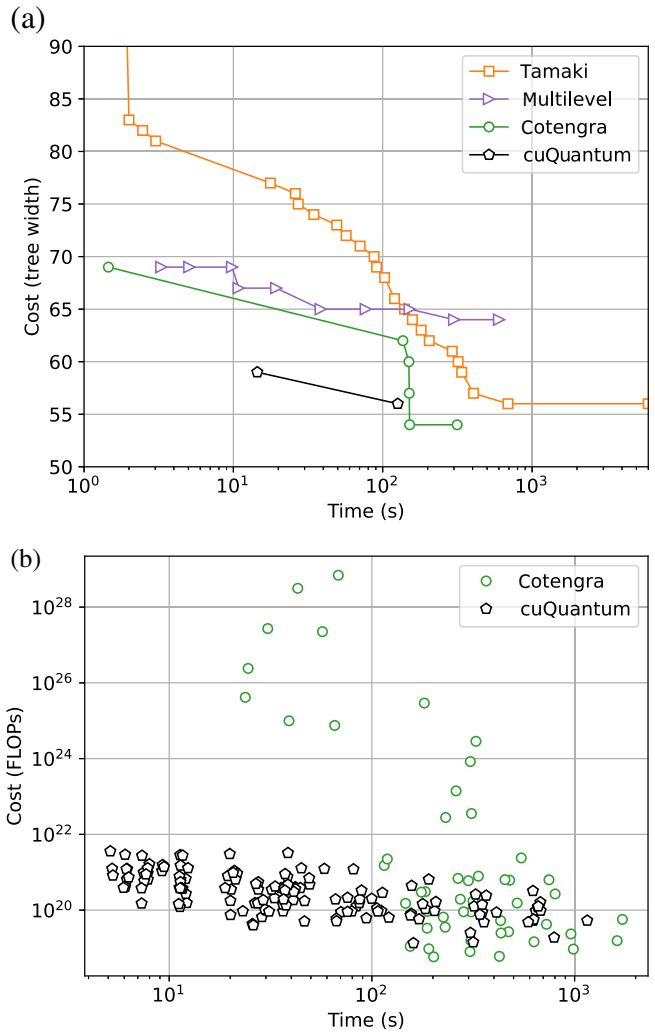


FIG. 31. Optimized contraction cost as a function of optimization time using a single CPU thread of an AMD EPYC "Milan" processor. (a) Tree width for a single quantum circuit without slicing. (b) FLOP count for ten quantum circuits with slicing.

To better compare cuQuantum and cotengra, we generate ten random $(N = 56, d = 10)$ circuit instances to capture random fluctuations in the optimized costs (due to both the differences in circuits and randomness in the optimizer). Both optimizers readily return the contraction cost in FLOP counts, which is reasonable proxy for the actual contraction time, and the results are shown in Fig. 31(b). We observe that, at very short times, cuQuantum is capable of producing good optimization results. On the other hand, while cotengra requires more time to produce good results, it is generally very competitive with cuQuantum if both are allowed $\gtrsim$100 s of optimization time.

APPENDIX I: ACHIEVABLE FIDELITIES WITH DMRG

Time evolving an approximate tensor network ansatz is a promising method to simulate quantum circuits with a limited fidelity. The most well-studied tensor networks in the context of quantum circuit simulations are matrix product states (MPS) and projected entanglement pair states (PEPS). For simulating transition amplitudes in random circuit sampling, the best-performing algorithm that we are aware of is the variant of time-dependent DMRG presented in Refs. [6,7], which is based around an approximate MPS representation of the full quantum state. Compared to exact simulation, the difficulty of approximate simulation—especially in the regime of submaximal entanglement—is much more difficult to reliably assess. This appendix elaborates on the methods used in Sec. V B of the manuscript and extends them by estimating how the cost of approximate simulation grows with circuit depth for RG circuits using a combination of DMRG and the bounds on achievable MPS fidelities described in Ref. [20].

An MPS representation of a quantum state can be chosen in many different ways, and we consider blocking of the qubits into b blocks $[\mathcal{S}_1, \ldots, \mathcal{S}_b]$ of size $[s_1, \ldots, s_b]$, where each set $\mathcal{S}_j$ contains the qubits in the associated block and is assigned to a single MPS tensor. In the event that all blocks are the same size, we refer to a given blocking scheme compactly as $b \times [s]$. We denote by $\mathcal{I}_j$ a superindex containing a single binary-valued index for each qubit in the associated block $\mathcal{S}_j$. Each block can then be assigned a tensor $M^{(j)}_{\alpha_j,\mathcal{I}_j,\beta_j}$, and the full quantum state is approximated as a contraction over these tensors as

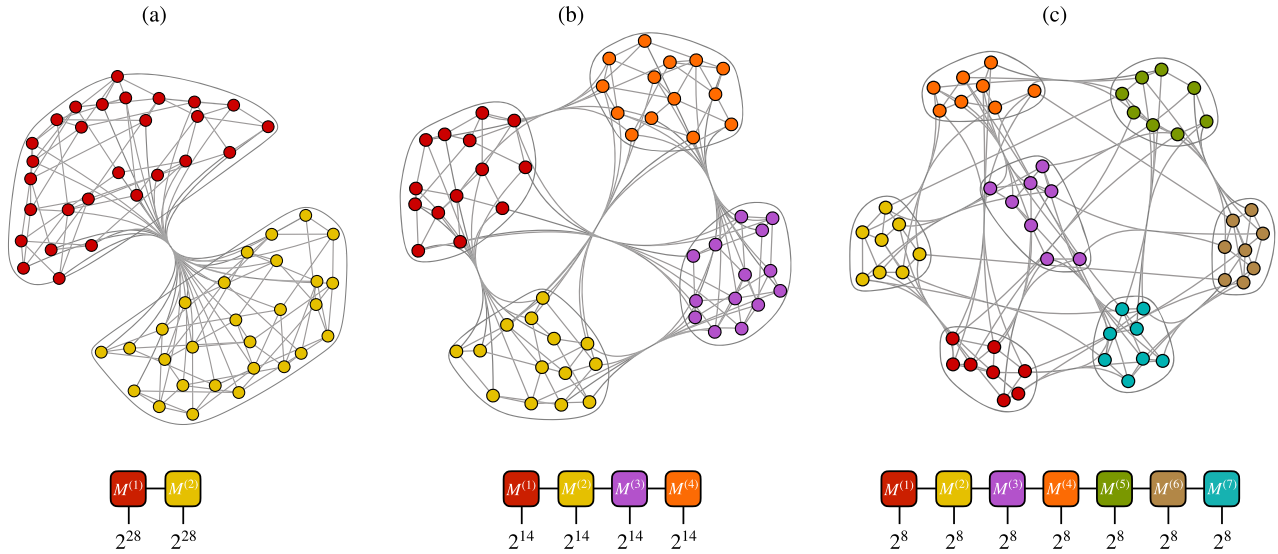


FIG. 32. Graph partitioning for blocked MPS simulations of a depth-5 circuit on 56 qubits. The graph from which the circuit is constructed (shown without edge coloring) is partitioned into $b = 2$ (a), $b = 4$ (b), or $b = 7$ (c) using KaHyPar. The objective is to minimize the number of interblock edges, which ultimately (upon assigning blocks to MPS tensors) minimizes the number of gates that must be applied between MPS tensors.

$$|\Psi\rangle = \sum_{\mathcal{I}_1,\ldots,\mathcal{I}_b} \sum_{\alpha_1=1}^{\chi} \cdots \sum_{\alpha_{b-1}=1}^{\chi} M^{(1)}_{\mathcal{I}_1,\alpha_1} M^{(2)}_{\alpha_1,\mathcal{I}_2,\alpha_2} \cdots \times M^{(b)}_{\alpha_{b-1}\mathcal{I}_b} |\mathcal{S}_1\rangle \otimes \cdots \otimes |\mathcal{S}_b\rangle. \tag{I1}$$

Here, χ is the bond dimension, which we assume to be the same for all tensors but, in general, need not be. An MPS of this form, and its relation to the blocking of qubits, is illustrated graphically for $b = 2, 4, 7$ in Figs. 32(a)–32(c). A blocked MPS can always be expanded into a more standard single-site MPS, with the blocked qubits corresponding to groups of adjacent single-site tensors. In this view, the blocking effectively allows the nominal interblock bond dimension χ to grow larger within the blocks; in the presence of inhomogeneous entanglement structure, blocking that is congruent with such structure can plausibly improve the fidelity of an MPS at fixed total resources. For a given blocking strategy $b \times [s]$, we assign qubits to blocks by using the open source hypergraph partitioning package KaHyPar to minimize the number of interblock $2Q$ gates. Examples of such partitionings and their correspondence with blocked MPS are shown in Fig. 32 for $b = 2, 4, 7$.

We say that a given quantum computer can be "classically spoofed" if, given a random circuit C, classical RCS can be achieved with comparable fidelity to that achieved by the quantum computer (e.g., as measured by the achievable XEB of samples) using a comparable amount of time per sample. One approach to spoofing would be to evolve an MPS through the circuit to approximate $|\Psi\rangle = C|0\rangle$ and then query the amplitudes of $|\Psi\rangle$ in the computational basis. While this approach has the benefit that sampling can be performed very directly, Ref. [7] advocated an alternate approach in which the circuit is split into lower depth circuits, for example, as $C = C_2 C_1$. By approximating (through MPS evolution) both $|\Psi_1\rangle = C_1|0\rangle$ and $|\Psi_2\rangle = C_2^\dagger|\boldsymbol{x}\rangle$, computational basis state amplitudes at the output of $\mathscr{C}$ can then be computed from the overlap of those states as

$$\langle \boldsymbol{x}|C_2 C_1|0\rangle = \langle \Psi_2|\Psi_1\rangle, \tag{I2}$$

and such calculations can then be used (e.g., in combination with rejection sampling) to draw samples from the output distribution of the circuit. An improved variation of this simulation scheme is to split the circuit into three pieces, as $C = C_2 C_M C_1$, computing amplitudes as expectation values $\langle \Psi_2|C_M|\Psi_1\rangle$. However, we have found that, for circuits with random geometries, having even a single layer of gates in C_M increases the contraction cost dramatically, and we have attempted simulations only with the circuit decomposed as in Eq. (I2).

The DMRG calculations reported in the paper can be used to estimate the required bond dimension to achieve comparable circuit fidelities to those achieved in the experiment. In particular, one can extrapolate the error per gate as a function of bond dimension to the $\varepsilon \approx 3.2 \times 10^{-3}$ value measured in the experiment for various blocking choices at different depths. Figure 33 shows such an extrapolation for circuits of depth 10 (where the distribution of output probabilities becomes well converged to Porter-Thomas *and* the exact contraction cost first becomes nearly saturated) and depth 20 (the deepest circuits for

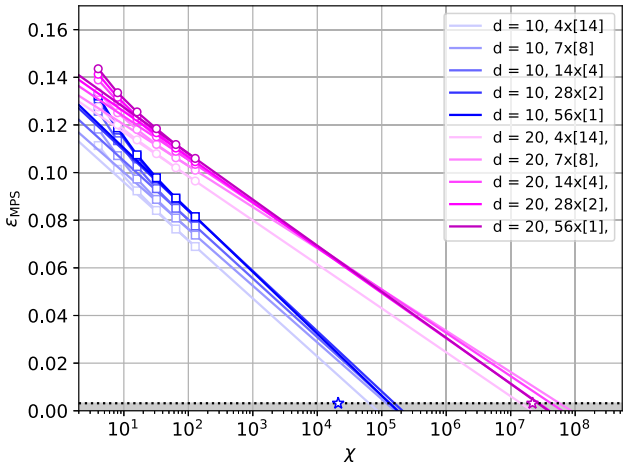


FIG. 33. Error per gate $\varepsilon_{\mathrm{MPS}}$ in DMRG simulations for depth-10 (blue) and depth-20 (magenta) circuits using a variety of blocking strategies. The lines are a linear fit to the largest two bond dimensions plotted for each blocking strategy ($\chi = 64, 128$) and provide estimates of the bond dimension required to achieve the experimental error rates (dotted horizontal line at the bottom of the plot). The stars are lower bounds on the required bond dimension for an optimal bipartite MPS extracted from Clifford simulations (see Fig. 35 and supporting text).

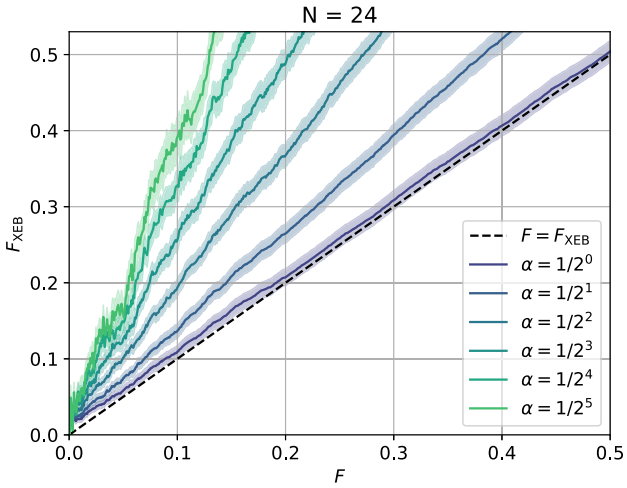


FIG. 34. Achievable $F_{\mathrm{XEB}}^{\mathrm{lin}}$ by sampling from bit strings whose amplitudes have been calculated using a closed MPS simulation at fidelity F. The various curves represent the achievable improvements by postprocessing the sampled bit strings, showing that fidelity improvements by postselecting on large probabilities require exponential postselection overhead (i.e., the ratio of $F_{\mathrm{XEB}}^{\mathrm{lin}}/F$ increases roughly linearly as the fraction of retained data, α, is decreased exponentially).

which we report experimental samples). We find only a weak dependence of the extrapolated bond dimension on the blocking choices and, in general, find that bond dimensions in excess of 10^5 (10^7) are likely necessary for classical spoofing of H2 via DMRG given circuits of depth 10 (20). Note that all of the blocking results lead to error per gate curves that are concave up, and extrapolation from relatively small bond dimensions may well bias the extrapolated bond dimension down relative to the true requirements.

In Ref. [20], the authors consider the bond dimension requirements to write down a bipartite MPS $|\Phi_{\mathrm{MPS}}\rangle$ that approximates an arbitrary state $|\Psi\rangle$ with fidelity $f = |\langle\Phi_{\mathrm{MPS}}|\Psi\rangle|^2$. In particular, they show that the fidelity is bounded above as $f^2 \leq \chi \mathrm{Tr}(\rho^2)$, where ρ is the reduced density matrix of the exact state $|\Psi\rangle$ on either side of the bipartition. Since the computation of amplitudes by evolving two MPSs toward the middle of the circuit achieves an effective simulation fidelity $F \sim f^2$ (being diminished by the loss of fidelity of both the forward and backward evolutions), they conclude that the simulation fidelity F achievable at fixed bond dimension is bounded above as $F \leq \chi \mathrm{Tr}(\rho^2)$.

Discrepant accounts of the relationship between the achievable MPS fidelity and the associated $F_{\mathrm{XEB}}^{\mathrm{lin}}$ have been reported in the literature (see Refs. [7,20]), though we note that the fidelities here are sufficiently large that the discrepancies between these two references are relatively inconsequential. We also perform numerical simulations for our circuits in the range of fidelities relevant to the experiment and find that $F = f^2$ serves as an excellent proxy for the achievable values of $F_{\mathrm{XEB}}^{\mathrm{lin}}$ from sampling based on closed-simulation amplitude calculations. For these calculations, we consider RG circuits on 24 qubits at depth 10 (at which point the output distribution of the exact circuits is well converged to the Porter-Thomas distribution). We sample a random bit string $\boldsymbol{x}$ and then evolve $|0\rangle$ (forward) and $|\boldsymbol{x}\rangle$ (backward) to the midpoint of the circuit. Both states are then optimally approximated by a bond dimension χ MPS via direct singular-value decomposition (SVD), and we record both the estimated probability for bit string $\boldsymbol{x}$ (squared overlap of the two MPSs) and the fidelity $F = f_0 f_{\boldsymbol{x}}$, where $f_{0(\boldsymbol{x})}$ is computed from the discarded weight of the SVD for the time-evolved $|0(\boldsymbol{x})\rangle$ state. We compute the amplitudes of 1000 randomly chosen bit strings for 100 random circuits (for a total of 100 000 amplitudes), and then for each circuit we estimate the achievable $F_{\mathrm{XEB}}^{\mathrm{lin}}$ by resampling from the associated list of 1000 estimated probabilities and computing $F_{\mathrm{XEB}}^{\mathrm{lin}}$ from those samples. In Fig. 34, the lowest (bluest, $\alpha = 1$) solid curve is a parametric plot of $F(\chi)$ versus $F_{\mathrm{XEB}}^{\mathrm{lin}}(\chi)$ averaged over the 100 sampled circuits (with the shaded region indicating the standard error on the mean over those circuits), showing excellent correlation. In general, various postprocessing strategies might be employed to increase the achievable $F_{\mathrm{XEB}}^{\mathrm{lin}}$ given the list of estimated probabilities. The other curves ($\alpha < 1$) in Fig. 34 are obtained by postprocessing the MPS estimates of the 1000 bit string

probabilities computed for each circuit by retaining just the largest $\alpha \times 1000$ largest samples (similar to the top-k postprocessing method described in Ref. [21]). While such postprocessing can produce $F_{\mathrm{XEB}}^{\mathrm{lin}}$ values larger than the achieved simulation fidelity F, the linear growth of $F_{\mathrm{XEB}}^{\mathrm{lin}}$ apparent in Fig. 34 as α is decreased exponentially suggests that the retained fraction of bit strings must shrink exponentially in the desired multiplicative enhancement of the fidelity. This behavior is expected for exact sampling given the exponential decay of the exact Porter-Thomas distribution, which makes sampling bit strings with probabilities R times larger than $1/2^N$ exponentially unlikely in R. We attribute the persistence of this behavior in approximate sampling to the distribution of approximated probabilities also having an exponential tail.

Given the exponentially large cost of enhancing $F_{\mathrm{XEB}}^{\mathrm{lin}}$ via postprocessing, we see no obvious better strategy to achieve (on average across circuits) a given XEB score $F_{\mathrm{XEB}}^{\mathrm{lin}}$ than simply choosing a bond dimension large enough that the MPS fidelity $F = f_0 f_x$ is comparable to $F_{\mathrm{XEB}}^{\mathrm{lin}}$. Returning to the analysis in Ref. [20] and the bound $F \leq \chi \mathrm{Tr}(\rho^2)$ derived therein, one should therefore use a bond dimension bounded below as

$$\chi \geq F/\overline{\mathrm{Tr}(\rho^2)}, \tag{I3}$$

where the overline denotes an average over circuits. As a result of the 2-design property of the Clifford group, $\overline{\mathrm{Tr}(\rho^2)}$ can be computed efficiently for our circuits by replacing the Haar random $1Q$ gates with i.i.d. random $1Q$ Clifford gates (see, for example, Ref. [74]) and then performing the average numerically using efficient Clifford simulations. Lower bounds on the bond dimension from such simulations are shown for $n \in \{16, 24, 32, 40, 48, 56\}$ in Fig. 35. For 56 qubits, this bound is also shown in Fig. 33 at depths $d = 10$, 20 (blue and magenta stars, respectively). In practice, a DMRG simulation may underperform relative to this bound for several reasons, including (i) the bound is tight only for states with a flat Schmidt spectrum, which will not be encountered for the non-Clifford circuits considered in this manuscript, and (ii) DMRG is a greedy algorithm that may not produce the globally optimal MPS. Indeed, we see that, for a variety of blocking strategies, extrapolation of the DMRG results to determine what bond dimension is required to achieve the experimentally measured fidelities indicates that the bound is *not* saturated in practice (at least in our implementation of the DMRG algorithm), requiring bond dimensions in the neighborhood of 10^5 even for depth-10 circuits. While it is not entirely clear how far the bond dimension in DMRG simulations could be pushed in practice, we note that the largest MPS-based simulation performed to date (by a large margin) used a bond dimension of $2^{16} = 65536$ [75]. Utilizing approximately 1000 tensor-processing-unit cores, this calculation took on the order of minutes for a single MPS optimization, of which many are required to simulate even a depth-10 circuit. While this time could surely be brought down to some extent, we note that the deeper circuits here require *much* larger bond dimensions (e.g., the magenta lines and star in Fig. 33). Moreover, slight modifications of the $1Q$-gate set can lower the depth at which the purity saturates to its Haar-random ($\sim 1/2^{N/2-1}$) value. For example, fixing the $1Q$ gates to be $\pi/2$ rotations about axes in the xy plane of the Bloch sphere (as in Ref. [20]) avoids situations—encountered early in circuits with the Haar-random SU(2) gates used here—in which the $1Q$ gates leave the $|0\rangle$ state close to the south or north pole of the Bloch sphere [which then reduces the amount of entanglement the $U_{ZZ}(\pi/2)$ gates can generate]. Simulations using a Clifford $1Q$-gate set restricted to such rotations yield the dotted line in Fig. 35, showing that even depth-12 circuits (the same depth at which the cost of memory-unconstrained contraction saturates) can require near-maximal bond dimensions.

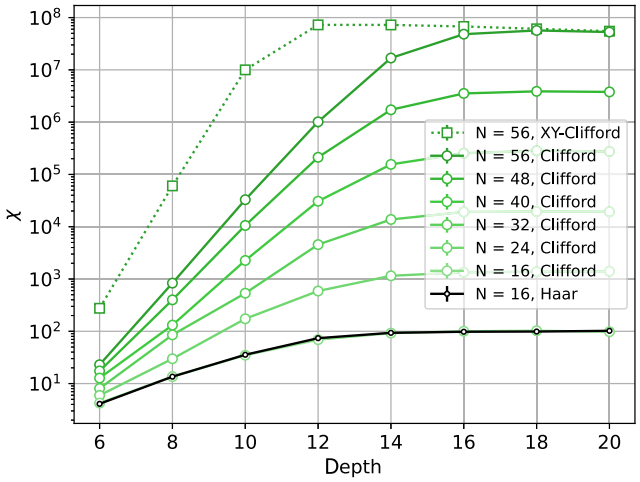


FIG. 35. Clifford simulations of the average bipartite purity [converted into a minimum bond dimension χ using Eq. (I3)] for our circuits at various system sizes. All curves are averaged over 100 random geometries and ten random assignments of $1Q$ gates per geometry, with the optimal blocking determined for each geometry independently. The black curve shows simulations with Haar-random $1Q$ gates for $N = 16$, confirming the equivalence with averages over Clifford circuits.

APPENDIX J: TABLE OF ACRONYMS

This appendix contains a list of all acronyms and variables used in this paper and their associated definitions (Table III).

TABLE III. Summary of acronyms and variables used in this work.

Label	Meaning
QCCD	Quantum charge-coupled device
QEC	Quantum error correction
RCS	Random circuit sampling
N	Number of qubits
SPAM	State preparation and measurement
RB	Randomized benchmarking
ϵ_{2Q}	Average two-qubit gate infidelity
ϵ_{mem}	Average memory error per qubit per layer
ϵ_{1Q}	Average one-qubit gate infidelity
p_{SPAM}	Average probability of SPAM error
$\varepsilon(N)$	Effective process infidelity per $2Q$-gate
$P(x)$	Ideal probability of measuring a bit string x
$\mathscr{P}(x)$	Empirical fraction of measured bit strings corresponding to x
XEB	Cross-entropy benchmark
MB	Mirror benchmarking
F	Fidelity
$F_{\mathrm{XEB}}^{\mathrm{lin}}$	Fidelity estimate from linear XEB
F_{GC}	Fidelity estimate from gate counting (digital error model)
F_{MB}	Fidelity estimate from mirror benchmarking
$F_{\mathrm{XEB}}^{\mathrm{log}}$	Fidelity estimate from log XEB
FLOP	Floating-point operation
TN	Tensor network
d	Circuit depth
D	Dimension of a spatially local circuit
$\mathscr{N}_{d,N}$	Effective qubit number
$\mathscr{C}_{d,N}$	Complexity density
$\mathscr{C}_{d}$	Asymptotic complexity density as $N \to \infty$
$\mathcal{G}_{d,N}$	Random d-regular graph on N nodes
$C_{d,N}$	Depth d circuit on N qubits generated from $\mathcal{G}_{d,N}$
RG	Random geometry
GPU	Graphics processing unit
M	Total number of samples in an RCS experiment
S	Number of samples per circuit in an RCS experiment
C	Number of different circuits in an RCS experiment
MPS	Matrix product state
DMRG	Density-matrix renormalization group
PEPS	Projected entanglement pair states
SVD	Singular-value decomposition
χ	Bond dimension of an MPS
$\varepsilon_{\mathrm{MPS}}$	Estimated error per $2Q$ gate in an MPS simulation
b	Number of blocks in a blocked MPS
s	Number of qubits per block in a blocked MPS
T_q	Total required quantum run-time of a verifiable RCS experiment
τ_q	Quantum run-time of a single shot of one circuit
$\mathscr{T}_C$	Tensor network associated to a unitary quantum circuit
$\mathscr{T}_C^0$	Tensor network $\mathscr{T}_C$ prior to Schmidt decomposition
r_j	Rank (number of legs) of a tensor $\mathscr{T}_j$

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